

Epistemic Closure Nets

Curvature, Holonomy, Certification, and Meta-Closure in an Expansive Network Formalism

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Abstract

This work unifies the SCE-IM closure framework [1]—windowed semantics, curvature gaps, zipper signatures, and stability—with the closed epistemic kernel: internal certification, theory atlases, holonomy obstructions, and meta-closure towers. The unification is network-expansive: a typed diagram of nodes (syntax, semantics, certificates, resources, refinements, and experimental harnesses) connected by morphisms and compatibility constraints. Claims are tagged by evidence type; non-closures are isolated as explicit conjectures; experimental nodes are specified with auditable estimators, uncertainty quantification, and pre-registered falsification patterns.

Contents

1	Scope, Contribution Summary, and Falsification Discipline	2
1.1	Scope	2
1.2	Auditability Rules	2
1.3	Main Contributions (Within This Paper)	3
2	Definitions (Nodes and Edges)	3
2.1	Evidence Contract	3
2.2	Kernel (Closure + Certification Layer)	4
2.3	SCE-IM / Epistemic Closure System with Metric Interface	4
2.4	Constraints as Geometry (CNF / cGCNF Nodes)	4
2.5	Finite-Bank Certification Node (Bank & Margin)	5
2.6	Protocol Holonomy (Typed Refinement Square; No Untyped Inverses)	5
2.7	Theory Atlas Node	6
2.8	The Network Object (Expansive, Non-Sequential)	7
2.9	Gauge Actions and Rigid Transport (Geometric Core)	7
3	Axioms (Consistency and Minimal Couplings)	8
4	Lemmas and Theorems (Local Compatibility and Obstructions)	9
5	Invariants (What Remains Under Equivalence)	11
6	Structural Predictions (Modeled, Falsifiable)	12
7	Declared Holes (Non-Closure) and Network Expansion	13
7.1	Conjectures as Safe Network Expansions	13

8	Network Diagram (Expansive)	14
9	Verifiable Extensions: Experimental Nodes and Protocols	15
9.1	Experimental Contract (Auditable Measurement)	15
9.2	Typed Protocol Holonomy: Deterministic vs History-Dependent Semantics	15
9.3	Monte Carlo Estimators for Window-Measures	16
9.4	Normalized Window Probabilities (Avoiding Explicit $\mu_{\Theta}(K)$)	16
9.5	Pre-Registration: Sample Size Planning and Decision Thresholds	17
9.6	Distances Between Predicted Distributions (for Atlas Probes)	17
9.7	Experimental Nodes (P1–P5) as Network Expansions	18
9.7.1	P1: Refinement-Square Holonomy Experiment (Finite-Bank)	18
9.7.2	P1*: Proof-of-Concept Artifact Node (Cut-State Holonomy)	18
9.7.3	P1**: Public-Data Instantiation (GW150914 Skeleton)	21
9.7.4	P2: Gray-Zone Scaling (Estimating Effective Dimension)	22
9.7.5	P3: Atlas Holonomy with Probes (Groupoid on Kernels)	23
9.7.6	P4: Zipper vs Protocol Holonomy Correlation Test (Bridge H1 as an Experiment)	23
9.7.7	P5: CNF-to-Volume Calibration Node	24
9.8	Pre-Registered Falsification Patterns (Diagnostics)	24
9.9	Pre-Registered Protocols (Decision Rules with Error Bars)	25
10	Physics/Tech Case Studies (End-to-End Harness Instantiations)	25
10.1	Case Study A: Materials Stability Under Competing Phases	26
10.2	Case Study B: Collider Analysis Pipeline	27
10.3	Case Study C: Gravitational-Wave Coherence Across Detectors	27
10.4	Reproducibility Bundle (Artifact Node)	28
10.5	Case Study D: Lattice QCD Systematics as Protocol Holonomy (Instantiation Template)	29
A	Provenance and Scope	29
B	Bibliography (Minimal Pointers)	29

1 Scope, Contribution Summary, and Falsification Discipline

1.1 Scope

This manuscript is not a completeness theorem for scientific knowledge. Its objective is an auditable specification layer that (i) separates soundness from completeness, (ii) makes finite-resource effects measurable, and (iii) isolates non-closures as typed obstructions.

1.2 Auditability Rules

- (A1) **Finite probes for ideal objects.** Every ideal object is paired with finite windows/banks used for certification.
- (A2) **Explicit quantifiers.** Every existential/universal claim fixes its domain (windows, banks, templates, or protocols).
- (A3) **Evidence typing.** Each claim is tagged [Proved], [Model], or [Conjecture].

- (A4) **Error bars.** Every numerical or statistical claim comes with an auditable uncertainty contract.
- (A5) **Pre-registered falsification.** For any empirical test node, the net (nodes/edges and decision rules) is frozen before sampling. Network expansion is allowed only as a separate post-mortem branch, never as a way to rescue a failed prediction.

Remark 1.1 (Notation hygiene: declared overloads). Some symbols are contextually overloaded and always disambiguated by their arguments or subscripts: E (SCE-IM instance of Definition 2.5 vs. certified-set functor of Definition 2.13 vs. certified sets $E_\tau, E_\tau^{(N)}, E(S, \tau)$); S (syntactic space of Definition 2.5 vs. template elements/banks of Axiom 3.6 and Sections 9–10); T (teeth of Definition 2.5 vs. charts of Section 2 vs. Monte Carlo sample size of Section 9); K (kernel K_r vs. window K vs. Markov kernel of Definition 9.24); F (kernel evolution vs. score functional vs. CNF formula); Φ (SCE-IM dynamics vs. cGCNF formula); Ω (semantic space vs. chart validity region $\Omega_r(T)$); I (literal scope vs. chart index set vs. threshold interval of Definition 5.1); τ (tooth in Axiom 3.3 vs. positive margin elsewhere); and Prob (measure sets vs. the Markov-kernel category of Definition 9.24).

1.3 Main Contributions (Within This Paper)

- (C1) A typed closure net connecting internal certification, windowed semantics, and refinement dynamics.
- (C2) A clean separation between index-trivial holonomy and program-level (history-dependent) holonomy.
- (C3) An atlas-transition layer whose triple-overlap cocycles obstruct global trivialization of predictions.
- (C4) A finite-bank transfer theorem derived from explicit continuity moduli and covering-number control (with empirical estimators for both).
- (C5) A suite of verifiable experimental nodes with auditable Monte Carlo estimators and pre-registered tests.
- (C6) A minimal artifact node proof-of-concept (Section 9.7.2) exhibiting the diagnostic regime $\hat{U} \rightarrow 0$ while $\hat{\Delta}_\square \not\rightarrow 0$ under a controlled hidden-state ξ , with a frozen pre-registration and a reproducibility artifact.

2 Definitions (Nodes and Edges)

2.1 Evidence Contract

Definition 2.1 (Evidence tags). Statements are typed as [Proved], [Model], [Conjecture], and the tag is part of the type system of the manuscript: it adds no content, but fixes what kind of audit (proof, model contract, or open problem) is being asserted. Unless explicitly stated otherwise, [Proved] means derivable from the axioms in Section 3 relative to standard metatheory (e.g. ZFC).

Remark 2.1 (Terminology: “curvature” and “holonomy”). In this paper, curvature means an operational gap (an infimum of an error functional under explicit resources, cf. [2]), not Riemannian sectional curvature. Similarly, holonomy is used in the minimal sense of path dependence under

typed refinements (protocol holonomy) or triple-overlap cocycles (atlas holonomy). Whenever a genuinely geometric structure (group action / cocycle class / conjugacy invariance) is present, it is stated explicitly.

2.2 Kernel (Closure + Certification Layer)

Definition 2.2 (Closed epistemic kernel). A closed epistemic kernel at resource $r \in \mathbb{N}$ is the tuple

$$K_r = (X, \sim, O_r, F, \Sigma^*, \text{supp}, \text{Cert}_r, \mathcal{J}_r),$$

where $X \neq \emptyset$, $F : X \rightarrow X$ is evolution/refinement, $\sim$ is gauge equivalence on X , O_r is a family of observables $o : X \rightarrow Y_o$ factoring through $X/\sim$, Σ^* is a token space, $\text{supp} : X \rightarrow \mathcal{P}(\Sigma^*)$ is available-token support, $\text{Cert}_r : X \times \Sigma^* \rightarrow \{0, 1\}$ is internal certification, and $\mathcal{J}_r : \Sigma^* \rightarrow \mathcal{P}(X)$ is a partial semantics (soundness only).

Definition 2.3 (Accessible observations). For $x \in X$, define

$$\text{Obs}_r(x) := \{(o, o(x)) : o \in O_r\}.$$

Definition 2.4 (Meta-certificate (certificate layer)). The pair $(\text{supp}, \text{Cert}_r)$ is the certification layer: it produces and validates tokens. No completeness is assumed.

2.3 SCE-IM / Epistemic Closure System with Metric Interface

Definition 2.5 (SCE-IM instance). An SCE-IM instance [1] is a tuple

$$E = (S, O, \Omega, \mathcal{T}, \mathcal{J}, \text{err}, \mu, K, \Phi),$$

where S is a syntactic space, O an objective space, Ω a semantic space, $\mathcal{T}$ a set of teeth (constraints), $\mathcal{J} : S \rightarrow \mathcal{P}(\Omega)$ a semantics map, $\text{err} : S \times O \rightarrow [0, \infty]$ an error functional, μ a measure on Ω , K a family of admissible windows (measurable subsets of Ω with $\mu(K) < \infty$), and Φ a dynamics on S .

Definition 2.6 (Windowed volume). For a window $K \in \mathcal{K}$ and $\sigma \in S$,

$$\text{Vol}_K(\sigma) := \mu(\mathcal{J}(\sigma) \cap K).$$

Definition 2.7 (Resource-limited curvature). Given a resource functional $\rho : S \rightarrow [0, \infty)$ and objective $o \in O$,

$$\kappa_R(o) := \inf\{\text{err}(\sigma, o) : \rho(\sigma) \leq R\}, \quad \kappa(o) := \inf_{\sigma \in S} \text{err}(\sigma, o).$$

2.4 Constraints as Geometry (CNF / cGCNF Nodes)

Definition 2.8 (Discrete CNF forbidden region). On $\{0, 1\}^n$, a CNF $F = \bigwedge_{j=1}^m C_j$ defines a forbidden region $U(F)$ as the union of clause-falsifying subcubes; the satisfying set is $\text{Mod}(F) = \{0, 1\}^n \setminus U(F)$.

Definition 2.9 (Continuous GCNF (cGCNF) [3]). To avoid collision with the kernel state space X of Definition 2.2, denote the ambient product space by $\mathcal{X}$. Let $\mathcal{X} = \prod_{i=1}^n X_i$ be a product of topological spaces. A literal is $\ell = (I, f, Y, U)$ with $I \subseteq [n]$, $f : \prod_{i \in I} X_i \rightarrow Y$ continuous, and $U \subseteq Y$ open. Define $\text{Mod}(\ell) = (f \circ \pi_I)^{-1}(U)$. A clause is a finite union of literals; a formula is a finite intersection of clauses. For a clause $C = \bigvee_{t=1}^m \ell_t$, define

$$\text{Forb}(C) := \bigcap_{t=1}^m (\mathcal{X} \setminus \text{Mod}(\ell_t)), \quad U(\Phi) := \bigcup_j \text{Forb}(C_j), \quad \text{Mod}(\Phi) := \mathcal{X} \setminus U(\Phi).$$

2.5 Finite-Bank Certification Node (Bank & Margin)

Definition 2.10 (Finite-bank transfer data [4]). Let $(\Theta, \mathcal{A}, \mu_\Theta)$ be a measure space and $K \in \mathcal{A}$ a window. Fix a margin parameter $\tau > 0$ and bank size N . Let $E_\tau \in \mathcal{A}$ be an “ideal” robust region and $E_\tau^{(N)} \in \mathcal{A}$ the region certified by a finite bank. Let ε_N be the bank covering radius and $\omega : [0, \infty) \rightarrow [0, \infty)$ a modulus. Under the regularity hypothesis of Axiom 3.6, the inclusions

$$E_{\tau+\omega(\varepsilon_N)} \subseteq E_\tau^{(N)} \subseteq E_\tau$$

are proved in Theorem 4.1.

2.6 Protocol Holonomy (Typed Refinement Square; No Untyped Inverses)

Definition 2.11 (Index category for refinements). Let $\mathcal{I}$ be a small category whose objects are pairs (N, τ) . For $N \leq N'$ there is a bank morphism

$$B_{N' \rightarrow N} : (N', \tau) \rightarrow (N, \tau),$$

(bank densification acts *contravariantly* on the index: it points from the larger bank to the smaller one), and for $\delta \geq 0$ there is a margin morphism

$$R_\delta : (N, \tau) \rightarrow (N, \tau + \delta).$$

Composition is defined by concatenation of refinements.

Definition 2.12 (Measured-set poset categories). Let $\text{Meas}^\subseteq(\Theta)$ be the poset-category of measurable subsets of Θ with a unique morphism $A \rightarrow B$ iff $A \subseteq B$. Let $\text{Meas}^\supseteq(\Theta)$ be the same objects with a unique morphism $A \rightarrow B$ iff $A \supseteq B$ (reverse inclusion).

Definition 2.13 (Certified-set functor (variance-aware)). A certified-set functor is a functor

$$E : \mathcal{I} \rightarrow \text{Meas}^\supseteq(\Theta), \quad E(N, \tau) := E_\tau^{(N)},$$

so that a refinement morphism maps to the corresponding inclusion: margin tightening points toward more restrictive certified sets, while bank densification (contravariant in the index) maps the bank morphism $B_{N' \rightarrow N}$ to the inclusion $E(N', \tau) \supseteq E(N, \tau)$ that the nested banks of Axiom 3.6(iv) provide. Equivalently: a morphism $(N, \tau) \rightarrow (N', \tau')$ in $\mathcal{I}$ implies

$$E_\tau^{(N)} \supseteq E_{\tau'}^{(N')}.$$

Definition 2.14 (Index-level square comparison (trivial holonomy)). Fix $N \leq N'$ and $\delta \geq 0$. There are two morphisms in $\mathcal{I}$ from (N', τ) to $(N, \tau + \delta)$:

$$p_1 := B_{N' \rightarrow N} \circ R_\delta, \quad p_2 := R_\delta \circ B_{N' \rightarrow N}.$$

Since E lands in a poset (Definition 2.12), the composite morphisms $E(p_1)$ and $E(p_2)$ both identify the same endpoint object $E(N, \tau + \delta)$. Therefore any “holonomy” defined purely at the index level is forced to be trivial. We record this by defining, for any measurable window $K \subseteq \Theta$,

$$\Delta_{\square}^{\text{idx}}(N, N', \tau, \delta; K) := 0.$$

Nontrivial protocol holonomy is defined at the program level (Definition 9.5) where refinement semantics can be history-dependent.

Remark 2.2 (Legacy shorthand). The earlier commutator notation $L := B^{-1}R_\delta^{-1}BR_\delta$ is retained as informal shorthand for “compare two refinement paths in a square”. It becomes literal only after upgrading refinements to isomorphisms in a localized groupoid of program states (Definition 9.3 defines the underlying program category; no such localization is constructed here), or after choosing non-canonical sections that provide partial inverses. At the index level, the square comparison is well-typed through the functor E of Definition 2.13 and holonomy is declared trivial by Definition 2.14.

2.7 Theory Atlas Node

Definition 2.15 (Observable output assignment and probability functor). For each observable $o \in O_r$, fix a measurable output space $(Y_o, \mathcal{B}_o)$. Regard O_r as a discrete category O_r^{disc} (objects are observables; only identity morphisms). Define the output assignment functor

$$Y : O_r^{\text{disc}} \rightarrow \mathbf{Meas}, \quad o \mapsto (Y_o, \mathcal{B}_o),$$

where $\mathbf{Meas}$ is the category of measurable spaces and measurable maps. Let

$$P : \mathbf{Meas} \rightarrow \mathbf{Set}, \quad (Y, \mathcal{B}) \mapsto \text{Prob}(Y)$$

be the probability-measure functor into sets (here $\text{Prob}(Y)$ denotes probability measures on $(Y, \mathcal{B})$). Then $(P \circ Y)(o) = \text{Prob}(Y_o)$.

Definition 2.16 (Prediction section (typed)). A chart T determines a prediction section

$$\text{Pred}_T \in \Gamma(P \circ Y) := \prod_{o \in O_r} \text{Prob}(Y_o),$$

i.e. an assignment $o \mapsto \text{Pred}_T(o) \in \text{Prob}(Y_o)$. When convenient, we also regard Pred_T as a choice of one element in each fiber of the functor $P \circ Y : O_r^{\text{disc}} \rightarrow \mathbf{Set}$ (a section in the categorical sense; the section space is the product $\Gamma(P \circ Y)$ because O_r^{disc} is discrete).

Definition 2.17 (Chart validity region). A chart T has a validity region $\Omega_r(T) \subseteq X$.

Definition 2.18 (Atlas of charts). An atlas at resource r is a family $\{T_i\}_{i \in I}$ together with a declared (pre-registered) intended domain $A_r \subseteq X$, such that the accessible regime is covered: $A_r \subseteq \bigcup_i \Omega_r(T_i)$.

Definition 2.19 (Transition operators (cochain structure)). On overlaps $\Omega_r(T_i) \cap \Omega_r(T_j)$ we assume a family of bijections

$$g_{ij} = (g_{ij,o})_{o \in O_r}, \quad g_{ij,o} : \text{Prob}(Y_o) \rightarrow \text{Prob}(Y_o),$$

with pointwise inverses $g_{ji,o} = g_{ij,o}^{-1}$. These operators act on prediction sections by

$$(g_{ij} \cdot \text{Pred})(o) := g_{ij,o}(\text{Pred}(o)).$$

The family $\{T_i, g_{ij}\}$ is a graph of transition operators over the chart set—equivalently, a 1-cochain for the cover $\{\Omega_r(T_i)\}$ with values in bijections of $\Gamma(P \circ Y)$ —and we write $\mathfrak{G}_r$ for it. It acquires an ordinary groupoid structure (with composition $g_{ik} := g_{ij} \circ g_{jk}$: translate first by g_{jk} , then by g_{ij} , i.e. g_{ij} reads chart- j quantities into chart i) *exactly* when all triple cocycles are trivial (Definition 2.20): the cocycle $h_{ijk} = g_{ij} \circ g_{jk} \circ g_{ki}$ is precisely the obstruction to path-independence $g_{ik} = g_{ij} \circ g_{jk}$.

Nontrivial holonomy is precisely the failure of that condition, so the correct structure in the presence of holonomy is 2-categorical (a gerbe-like datum; cf. Conjecture 7.3 and [9, 10]). If O_r is upgraded from discrete to a category with nontrivial morphisms, impose the naturality constraint; in the default discrete case, naturality is vacuous. *Realization note:* each $g_{ij,o}$ may be induced by an invertible Markov kernel on Y_o ; this links to the kernel pushforward action in Definition 9.24.

Definition 2.20 (Atlas holonomy cocycle). On triple overlaps define the cocycle operator

$$h_{ijk} := g_{ij} \circ g_{jk} \circ g_{ki},$$

meaning componentwise, for each $o \in O_r$,

$$h_{ijk,o} := g_{ij,o} \circ g_{jk,o} \circ g_{ki,o} : \text{Prob}(Y_o) \rightarrow \text{Prob}(Y_o).$$

Thus h_{ijk} is an automorphism of the section space $\Gamma(P \circ Y)$ via pointwise action. If some $h_{ijk,o} \neq \text{id}$ (in the chosen gauge quotient), the atlas has nontrivial holonomy.

Remark 2.3 (Gauge dependence and conjugacy invariance). Changing the representative transitions by a 0-cochain $\{a_i \in \mathcal{G}_r\}$ (Definition 2.22) via $g'_{ij} := a_i g_{ij} a_j^{-1}$ conjugates the cocycles: $h'_{ijk} = a_i h_{ijk} a_i^{-1}$. Hence the conjugacy class of $\{h_{ijk}\}$ is the gauge-invariant holonomy datum. Whether this class is represented as a cohomology class on the nerve is the content of Conjecture 7.3.

2.8 The Network Object (Expansive, Non-Sequential)

Definition 2.21 (Closure net as a typed diagram). A closure net $\mathcal{N}$ is a typed directed multigraph (equivalently, a small category with decorations) whose nodes include:

{epistemic kernel K_r , SCE-IM instance (Definition 2.5), CNF/cGCNF, certified-set functor E (Definition 2.13), A

and whose edges are declared morphisms between them (semantics, volume, certification, refinements, transitions). A network extension adds nodes/edges while preserving all previously declared typing and compatibility constraints.

2.9 Gauge Actions and Rigid Transport (Geometric Core)

Definition 2.22 (Net-derived gauge group on predictions). Fix r and write the section space of predictions as

$$\Gamma_r := \Gamma(P \circ Y) = \prod_{o \in O_r} \text{Prob}(Y_o).$$

Let $\text{Aut}_{\text{rig}}(\mathcal{N}_r)$ denote the group of rigid closure-net automorphisms at resource r : rigid isomorphisms (Definition 2.23) from $\mathcal{N}_r$ to itself preserving the kernel data $(X, \sim, O_r, F, \Sigma^*, \text{supp}, \text{Cert}_r, \mathcal{I}_r)$, the certified-set functor E of Definition 2.13, and the atlas transition graph $\mathfrak{G}_r$; Definition 2.23 applied fiberwise on the observational family supplies the output-space bijections $\varphi_o : Y_o \rightarrow Y_o$ used below. Each $\varphi \in \text{Aut}_{\text{rig}}(\mathcal{N}_r)$ induces a componentwise pushforward

$$\rho_r(\varphi) : \Gamma_r \rightarrow \Gamma_r, \quad (\nu_o)_{o \in O_r} \mapsto (\varphi_o \# \nu_o)_{o \in O_r},$$

where $\varphi_o : Y_o \rightarrow Y_o$ are the measurable output-space bijections supplied by Definition 2.23. The canonical gauge group is the image

$$\mathcal{G}_r := \rho_r(\text{Aut}_{\text{rig}}(\mathcal{N}_r)) \leq \text{Aut}(\Gamma_r).$$

A choice of transition representatives $\{g_{ij}\}$ is a $\mathcal{G}_r$ -valued 1-cochain on the nerve of the cover $\{\Omega_r(T_i)\}$ (Definition 2.19).

Proposition 2.1 ([Proved] Rigidity $\Rightarrow$ gauge action). *The assignment $\rho_r : \text{Aut}_{\text{rig}}(\mathcal{N}_r) \rightarrow \text{Aut}(\Gamma_r)$ is a group homomorphism. Consequently, atlas-holonomy conjugacy classes computed from transition cochains are intrinsic to the rigid transport structure.*

Proof. Functoriality of pushforward under composition yields $\rho_r(\varphi \circ \psi) = \rho_r(\varphi) \circ \rho_r(\psi)$ and $\rho_r(\text{id}) = \text{id}$. Conjugacy invariance is the standard gauge change-of-trivialization calculation (Proposition 4.1). $\square$

Definition 2.23 (Rigid isomorphism of closure nets). Let E and E' be two instantiations (possibly on different spaces). A rigid isomorphism is a triple (h, g, u) transporting state space, objective space, and micro-space: $h : S \rightarrow S'$ and $g : O \rightarrow O'$ are bijections and $u : \Omega \rightarrow \Omega'$ is a measure-preserving bijection on the relevant σ -algebras, together with induced transports of windows, dynamics, and observables, such that error and windowed volumes are preserved. The observable transport comes equipped with measurable bijections $\varphi_o : Y_o \rightarrow Y_{g(o)}$ on output spaces commuting with the observables ($\varphi_o \circ o = g(o) \circ h$); for an automorphism of a fixed net with fixed observational family these are automorphisms $\varphi_o : Y_o \rightarrow Y_o$, and it is this fiberwise data that defines the pushforward action of Definition 2.22. (When instantiated from SCE-IM, this matches the standard “rigid isomorphism” notion used to prove curvature invariance.)

3 Axioms (Consistency and Minimal Couplings)

Axiom 3.1 (Gauge invariance). All observables $o \in O_r$ factor through $X/\sim$.

Axiom 3.2 (Soundness-only internal certification). For each r , there exists a partial semantics $\mathcal{J}_r$ such that

$$\text{Cert}_r(x, s) = 1 \implies x \in \mathcal{J}_r(s).$$

No completeness or totality is assumed.

Axiom 3.3 (Zipper refinement operator). There exists an operator $\triangleleft : S \times \mathcal{T} \rightarrow S$ and measurable sets $D_\tau \subseteq \Omega$, the declared semantic constraint region of each tooth τ (inert under the current axioms; consumed by the conjectural monoid layer, H2), such that

$$\mathcal{J}(\sigma \triangleleft \tau) \subseteq \mathcal{J}(\sigma) \cap D_\tau.$$

Axiom 3.4 (Lower semicontinuity of error). For each $o \in O$, the map $\sigma \mapsto \text{err}(\sigma, o)$ is lower semicontinuous on S .

Axiom 3.5 (Windowed measurability). For each $\sigma \in S$ and $K \in \mathcal{K}$, $\mathcal{J}(\sigma) \cap K$ is measurable and $\mu(K) < \infty$.

Axiom 3.6 (Bank regularity for transfer). There is a compact template space $(S_\infty, \text{dist}_S)$ and a score functional

$$F : \Theta \times S_\infty \rightarrow \mathbb{R}$$

such that:

- (i) for each $S \in S_\infty$ the map $\theta \mapsto F(\theta, S)$ is measurable;
- (ii) for each θ the map $S \mapsto F(\theta, S)$ is uniformly continuous with a common modulus of continuity ω (independent of θ), taken non-decreasing with $\omega(0) = 0$ (replace ω by its optimal, monotonized modulus if necessary):

$$|F(\theta, S) - F(\theta, S')| \leq \omega(\text{dist}_S(S, S'));$$

(iii) for each $\tau > 0$, the level-existence set $E_\tau := \{\theta \in \Theta : \exists S \in S_\infty, F(\theta, S) < -\tau\}$ is measurable (this holds, e.g., when Θ is a standard Borel space, $\mathcal{A}$ is its Borel σ -algebra, and F is jointly measurable with S_∞ a compact metric space: the projection is then analytic and hence universally measurable; in general we record measurability as a datum, cf. Definition 2.10).

(iv) the banks are nested: $S_N \subseteq S_{N'}$ whenever $N \leq N'$ (and correspondingly $\varepsilon_N \geq \varepsilon_{N'}$).

A finite bank $S_N \subset S_\infty$ is an ε_N -net: for every $S \in S_\infty$ there exists $S' \in S_N$ with $\text{dist}_S(S, S') \leq \varepsilon_N$.

Axiom 3.7 (Refinement typing). Refinements act as morphisms in the index category $\mathcal{I}$ (Definition 2.11) and induce monotone maps via the certified-set functor E (Definition 2.13); bank densification acts contravariantly on the index.

Axiom 3.8 (Atlas transition typing). Transitions form a family of bijections with pointwise inverses between prediction sections on overlaps (Definition 2.19). Atlas holonomy is defined by the triple-overlap cocycles (Definition 2.20).

Axiom 3.9 (Evidence separation). No axiom asserts: (i) global completeness of internal certification, (ii) that zipper signatures determine atlas/protocol holonomy without extra hypotheses, (iii) that holonomy must vanish.

4 Lemmas and Theorems (Local Compatibility and Obstructions)

Lemma 4.1 ([Proved] Monotonicity of resource-limited curvature). *If $R \leq R'$, then $\kappa_{R'}(o) \leq \kappa_R(o)$ for all $o \in O$.*

Proof. The feasible set $\{\sigma : \rho(\sigma) \leq R\}$ is contained in $\{\sigma : \rho(\sigma) \leq R'\}$; taking infima yields the claim. $\square$

Lemma 4.2 ([Proved] Openness of cGCNF model sets). *In cGCNF, $\text{Mod}(\Phi)$ is open in $\mathcal{X}$.*

Proof. Each literal model set is open as a preimage of an open set under a continuous map. Finite unions/intersections preserve openness. $\square$

Theorem 4.1 ([Proved] Finite-bank transfer from a modulus). *Assume Axiom 3.6. Fix $\tau > 0$ and define the ideal robust existence set and its banked version by*

$$E_\tau := \{\theta \in \Theta : \exists S \in S_\infty \text{ with } F(\theta, S) < -\tau\}, \quad E_\tau^{(N)} := \{\theta \in \Theta : \exists S \in S_N \text{ with } F(\theta, S) < -\tau\}.$$

Then the transfer inclusions hold:

$$E_{\tau+\omega(\varepsilon_N)} \subseteq E_\tau^{(N)} \subseteq E_\tau.$$

Proof. The inclusion $E_\tau^{(N)} \subseteq E_\tau$ is immediate since $S_N \subseteq S_\infty$. For the other inclusion, take $\theta \in E_{\tau+\omega(\varepsilon_N)}$. Then there exists $S \in S_\infty$ with $F(\theta, S) < -(\tau + \omega(\varepsilon_N))$. By ε_N -net coverage, pick $S' \in S_N$ with $\text{dist}_S(S, S') \leq \varepsilon_N$. By the modulus inequality and monotonicity of ω (Axiom 3.6(ii)),

$$F(\theta, S') \leq F(\theta, S) + \omega(\text{dist}_S(S, S')) \leq F(\theta, S) + \omega(\varepsilon_N) < -\tau,$$

so $\theta \in E_\tau^{(N)}$. $\square$

Corollary 4.1 ([Proved] Lipschitz modulus). *If, in addition, $S \mapsto F(\theta, S)$ is L -Lipschitz uniformly in θ , then one may take $\omega(\varepsilon) = L\varepsilon$.*

Proof. Immediate from the Lipschitz bound. $\square$

Theorem 4.2 ([Proved] Non-contradiction of epistemic kernel and SCE-IM). *Under Axioms 3.1–3.9, the combined system is consistent relative to the metatheory: a model exists in which the epistemic kernel and SCE-IM components coexist without forcing contradictions.*

Proof. We exhibit an explicit model of all of Axioms 3.1–3.9. *Kernel:* $X = \{x_0\}$, $\sim$ trivial, $O_r = \{o\}$ with $Y_o = \{0\}$ and $o(x_0) = 0$ (Axiom 3.1 holds), $F = \text{id}_X$, $\Sigma^* = \{s\}$, $\text{supp}(x_0) = \emptyset$, $\text{Cert}_r \equiv 0$ and $\mathcal{J}_r(s) = \{x_0\}$ (soundness holds vacuously: no token is ever certified, Axiom 3.2). *SCE-IM:* $S = \{\sigma_0\}$, $O = \{o_0\}$, $\Omega = \{\omega_0\}$, $\mathcal{T} = \emptyset$, $\mathcal{J}(\sigma_0) = \{\omega_0\}$, $\text{err} \equiv 0$ (lower semicontinuous, Axiom 3.4), μ the counting measure on Ω with $\mu(\Omega) = 1 < \infty$, $K = \{\Omega\}$ (Axiom 3.5), $\Phi = \text{id}_S$. The zipper is vacuous since $\mathcal{T} = \emptyset$ (Axiom 3.3). *Bank data:* $\Theta = \{\theta_0\}$, $S_\infty = \{S_0\}$ compact with the discrete metric, $F(\theta_0, S_0) = 0$, $\omega \equiv 0$ (non-decreasing with $\omega(0) = 0$), $S_N = \{S_0\}$, $\varepsilon_N = 0$, and $E_\tau = \{\theta : 0 < -\tau\} = \emptyset$ for every $\tau > 0$ (Axiom 3.6, measurability trivial). *Refinement typing:* by Definition 2.13, $E(N, \tau) = E_\tau^{(N)} = \emptyset$ for all (N, τ) , so E is the constant functor into $\text{Meas}^\supset(\Theta)$: every morphism maps to the (unique) morphism $\emptyset \rightarrow \emptyset$ (Axiom 3.7). *Atlas:* a single chart T with $\Omega_r(T) = X$, transition $g_{11} = \text{id}$, trivial cocycles (Axiom 3.8). Axiom 3.9 is vacuously satisfied. All sorts are finite sets, so the model exists in ZFC. $\square$

Theorem 4.3 ([Proved] Protocol holonomy: index-triviality and program non-commutativity). (Index level) *Under Definitions 2.12–2.14, the index-level quantity $\Delta_\odot^{\text{idx}}(N, N', \tau, \delta; K)$ is identically 0.* (Program level) *Fix a start program state c and two protocols $p_1, p_2 : c \rightarrow c_1, c_2$ in $\mathcal{I}_{\text{prog}}$ (Definition 9.3), and a window $K \subseteq \Theta$. Let $\Delta_\odot(p_1, p_2; K)$ be as in Definition 9.5. Then $\Delta_\odot(p_1, p_2; K) = 0$ iff the produced certified sets agree μ_Θ -a.e. on K , and $\Delta_\odot(p_1, p_2; K) > 0$ implies they differ on a μ_Θ -non-null subset of K .*

Proof. (Index level) By Definition 2.13, E lands in the poset $\text{Meas}^\supset(\Theta)$ (Definition 2.12), where any two parallel morphisms with the same source and target coincide; hence $E(p_1) = E(p_2)$ for the two paths of Definition 2.14, and $\Delta_\odot^{\text{idx}} \equiv 0$ is then the declared value of Definition 2.14. (Program level) Immediate from Definition 9.5: $\Delta_\odot(p_1, p_2; K) = \mu_\Theta((E_{p_1} \triangle E_{p_2}) \cap K)$ equals 0 exactly when the certified sets agree μ_Θ -a.e. on K , and is positive exactly when they differ on a set of positive μ_Θ -measure. $\square$

Theorem 4.4 ([Proved] Atlas holonomy obstructs global trivialization). *Under Axiom 3.8, say the atlas admits a global trivialization over the covered regime if there are automorphisms η_i of the section space Γ_r such that the descent condition*

$$g_{ij} = \eta_i^{-1} \eta_j \quad \text{for all } i, j$$

holds (equivalently: a choice of gauge $a_i := \eta_i$ in which all transitions are identities). If some cocycle $h_{ijk} \neq \text{id}$ in the gauge quotient, then no global trivialization exists.

Proof. If $g_{ij} = \eta_i^{-1} \eta_j$ for all i, j , then

$$h_{ijk} = g_{ij} g_{jk} g_{ki} = \eta_i^{-1} \eta_j \eta_j^{-1} \eta_k \eta_k^{-1} \eta_i = \text{id}.$$

By Proposition 4.1 the condition $h_{ijk} = \text{id}$ is gauge-invariant (only id is conjugate to id). Contrapositive: a nontrivial cocycle obstructs any such choice. $\square$

Proposition 4.1 ([Proved] Gauge invariance of atlas holonomy). *Under Definition 2.22 and the gauge update $g'_{ij} := a_i g_{ij} a_j^{-1}$, the cocycles transform by conjugation and the conjugacy class of the holonomy data is invariant. (If a cohomology theory on the nerve exists—Conjecture 7.3—then the represented cohomology class is likewise invariant; that conditional statement is not proved here.)*

Proof. Compute:

$$h'_{ijk} = g'_{ij} g'_{jk} g'_{ki} = (a_i g_{ij} a_j^{-1}) (a_j g_{jk} a_k^{-1}) (a_k g_{ki} a_i^{-1}) = a_i g_{ij} g_{jk} g_{ki} a_i^{-1} = a_i h_{ijk} a_i^{-1}.$$

Hence $h'_{ijk} = a_i h_{ijk} a_i^{-1}$, so the cocycles are conjugated by a_i and the conjugacy class is gauge-invariant. $\square$

Proposition 4.2 ([Proved] Curvature invariance under rigid transport). *Let (h, g, u) be a rigid isomorphism (Definition 2.23) transporting error as $\text{err}'(h(\sigma), g(o)) = \text{err}(\sigma, o)$. Then $\kappa'(g(o)) = \kappa(o)$ for all objectives o .*

Proof. Since h is a bijection (Definition 2.23), take infima over $\sigma' = h(\sigma)$:

$$\kappa'(g(o)) = \inf_{\sigma' \in S'} \text{err}'(\sigma', g(o)) = \inf_{\sigma \in S} \text{err}'(h(\sigma), g(o)) = \inf_{\sigma \in S} \text{err}(\sigma, o) = \kappa(o).$$

$\square$

Theorem 4.5 ([Proved] Soundness-only certification does not force a diagonal obstruction). *Soundness-only internal certification (Axiom 3.2) does not by itself imply a Gödel diagonal obstruction.*

Proof. A non-implication is established by exhibiting a model of the axioms in which no diagonal obstruction arises. Take the model constructed in the proof of Theorem 4.2, in which $\text{Cert}_r \equiv 0$: no token is ever certified, so in particular no self-referential sentence is certified and no diagonal phenomenon occurs. Hence Axiom 3.2 (which is satisfied there) does not by itself imply a diagonal obstruction. Obtaining such an obstruction requires *additional* structure: an internal encoding of sentences into tokens together with a reflection principle connecting Cert_r to semantic truth/representability (arithmetization is the classical sufficient route, not a necessary one). $\square$

5 Invariants (What Remains Under Equivalence)

Definition 5.1 (Zipper signature). Fix an objective $o \in O$ and an interval $I = [\varepsilon_0, \varepsilon_1]$ of sublevel thresholds for the error functional $\sigma \mapsto \text{err}(\sigma, o)$. For $\varepsilon \in I$ write $S_\varepsilon(o) := \{\sigma \in S : \text{err}(\sigma, o) \leq \varepsilon\}$ (closed by lower semicontinuity, Axiom 3.4, in the topology of S). The *merge tree* $\text{MT}_I(o)$ is the merge tree (dendrogram) of the filtration $\{S_\varepsilon(o)\}_{\varepsilon \in I}$ under the inclusions $S_\varepsilon(o) \subseteq S_{\varepsilon'}(o)$ for $\varepsilon \leq \varepsilon'$: its nodes are the connected components of the $S_\varepsilon(o)$ as they are born or merged while ε increases through I , labeled by their birth/merge heights. The *hitting-time observable* is $\tau^o(\sigma) := \inf\{\varepsilon \in I : \sigma \in S_\varepsilon(o)\}$, with $\inf \emptyset := +\infty$ (so that $\tau^o(\sigma) \geq \text{err}(\sigma, o)$). The zipper signature is

$$\Sigma_{\text{zip}}(E, o) := (\kappa(o), \text{MT}_I(o), \tau^o).$$

Definition 5.2 (Atlas holonomy invariant). The gauge-equivalence class of cocycles $\{h_{ijk}\}$ defines $\text{Hol}^{\text{atlas}}$.

Definition 5.3 (Protocol holonomy invariant). The family of square-comparison magnitudes $\Delta_{\square}(N, N', \tau, \delta; K)$ defines Hol^{prot} ; here the four-argument form abbreviates the program-level magnitude of Definition 9.5 evaluated on the canonical pair of protocols of Definition 9.21, i.e. $\Delta_{\square}(N, N', \tau, \delta; K) := \Delta_{\square}(p_1, p_2; K)$ for the square (N, N', τ, δ) , instantiated as follows: the start state is $c_0 = (S_N, \tau, \xi_0)$ with S_N a fixed pre-registered bank instance of size N and ξ_0 a fixed auxiliary state, the augmentation count is $m := N' - N$ for $N' \geq N$, and the two protocols are $p_1 := \text{Aug}^m \circ \text{Tighten}_\delta$ (tighten, then augment) and $p_2 := \text{Tighten}_\delta \circ \text{Aug}^m$ (augment, then tighten), both reaching bank size N' and margin $\tau + \delta$.

Proposition 5.1 ([Proved] Independence under current axioms). *Under Axioms 3.1–3.9, Σ_{zip} and $(\text{Hol}^{\text{atlas}}, \text{Hol}^{\text{prot}})$ are independent invariants: no implication between them is provable without extra coupling axioms.*

Proof. A non-implication in each direction is witnessed by a pair of models.

(Σ_{zip} does not determine holonomy.) Both models use the SCE-IM component of the proof of Theorem 4.2 (hence identical Σ_{zip} : $\kappa = 0$, trivial merge tree, trivial hitting times). Model (a) uses the single-chart trivial atlas of Theorem 4.2 (all $h_{ijk} = \text{id}$) and no refinement programs, so $(\text{Hol}^{\text{atlas}}, \text{Hol}^{\text{prot}})$ is trivial. Model (b) replaces the kernel by $X = \{x_0\}$ covered by three charts T_1, T_2, T_3 with $Y_o = \{0, 1\}$, $\text{Prob}(Y_o) \cong [0, 1]$; take $g_{12} = g_{23} = \text{id}$ and g_{31} the flip $p \mapsto 1 - p$ (with inverses $g_{21} = g_{32} = \text{id}$, g_{13} the flip, satisfying Definition 2.19; naturality is vacuous over the discrete observable category). Then $h_{123} = g_{12}g_{23}g_{31} = \text{flip} \neq \text{id}$ and the flip is not conjugate to id (conjugation is injective and fixes id : $aha^{-1} = \text{id}$ forces $h = \text{id}$; cf. Proposition 4.1), so $\text{Hol}^{\text{atlas}}$ is nontrivial while Σ_{zip} is unchanged. All other axioms are verified as in Theorem 4.2.

(Holonomy does not determine Σ_{zip} .) Keep the trivial holonomy structure of Model (a) but change the SCE-IM error to $\text{err}(\sigma_0, o_0) = 1$. Then $\kappa(o_0) = 1 \neq 0$, so Σ_{zip} changes while $(\text{Hol}^{\text{atlas}}, \text{Hol}^{\text{prot}})$ stays trivial.

Hence neither implication is derivable from Axioms 3.1–3.9 alone (Axiom 3.9 notwithstanding, which merely records this non-coupling at the presentation level). $\square$

6 Structural Predictions (Modeled, Falsifiable)

Proposition 6.1 ([Model] Gray-zone scaling with bank density). *Assume $\omega(\varepsilon) \approx L\varepsilon$ and $\varepsilon_N \asymp N^{-1/d}$ for an effective dimension d (estimated in the pool model by Definition 9.23). Then the bank-induced uncertainty thickness scales as $O(N^{-1/d})$ (for fixed margin τ and window K), under the band-regularity hypothesis of Proposition 9.3.*

Proposition 6.2 ([Model] Holonomy scaling with transfer modulus). *Assume that non-commutativity arises only through the bank-transfer gap measured by $\omega(\varepsilon_N)$. Then $\Delta_{\square}(N, N', \tau, \delta; K) = O(\omega(\varepsilon_N))$ as $N \rightarrow \infty$ with $N' \geq N$ along a pre-registered sweep and (τ, δ, K) fixed (e.g. with the gap $N' - N$ bounded, so that $\varepsilon_N \rightarrow 0$).*

Proposition 6.3 ([Proved] Atlas obstructions manifest as path dependence). *If $\text{Hol}^{\text{atlas}} \neq 0$, then there exist a triple of overlapping charts, an observable, and a prediction section such that translating the section from one chart to another along two different chart-paths (the direct transition versus the path through the third overlapping chart) yields different outputs.*

Proof. If $\text{Hol}^{\text{atlas}} \neq 0$, some cocycle satisfies $h_{ijk} \neq \text{id}$ (Definition 2.20; the condition is gauge-invariant by Proposition 4.1), so for some observable $o \in O_r$ the maps $g_{ij,o} \circ g_{jk,o}$ and $g_{ik,o}$ on $\text{Prob}(Y_o)$ differ (composition convention of Definition 2.19, using $g_{ik,o} = g_{ki,o}^{-1}$). Pick $\nu \in \text{Prob}(Y_o)$ with $(g_{ij,o} \circ g_{jk,o})(\nu) \neq g_{ik,o}(\nu)$ and extend it to a prediction section $\text{Pred} \in \Gamma_r$ (Definition 2.16). Translating Pred from chart k to chart i directly (by g_{ik}) versus through chart j (by $g_{ij} \circ g_{jk}$) yields path-dependent outputs at component o . $\square$

Definition 6.1 (Asymptotic trivialization in the resource limit). Let $r \mapsto c_r$ be a directed resource schedule (refinements in $\mathcal{I}_{\text{prog}}$). For a fixed loop type, i.e. a fixed protocol square as in Definition 9.21 issued from the state c_r , write $\Delta_{\square}(c_r; K) := \Delta_{\square}(p_1(r), p_2(r); K)$. Protocol holonomy is asymptotically trivial on window K if for every fixed loop type,

$$\lim_{r \rightarrow \infty} \Delta_{\square}(c_r; K) = 0.$$

Proposition 6.4 ([Model] Sufficient condition for $\Delta_{\mathcal{Q}} \rightarrow 0$). *If (i) bank transfer moduli satisfy $\omega(\varepsilon_r) \rightarrow 0$, (ii) the auxiliary state stabilizes ($\xi_r = \xi_{\infty}$ eventually), and (iii) atlas cocycle classes vanish on the relevant loop family, then $\Delta_{\mathcal{Q}}(c_r; K) \rightarrow 0$ as $r \rightarrow \infty$.*

Proposition 6.5 ([Model] Persistent holonomy decomposition). *In the same setting, observing $\hat{G} \rightarrow 0$ while $\hat{\Delta}_{\mathcal{Q}} \not\rightarrow 0$ supports one of: (a) non-stabilization of ξ (history-dependent semantics), (b) nontrivial atlas holonomy class (groupoid obstruction), (c) misspecified transfer modulus (failure of the tested regularity hypothesis). The artifact node in Section 9.7.2 realizes this persistent-holonomy pattern via a fixed window-choice state, in uncertainty-proxy form $\hat{U} \rightarrow 0$ while $\hat{\Delta}_{\mathcal{Q}} \not\rightarrow 0$.*

7 Declared Holes (Non-Closure) and Network Expansion

Definition 7.1 (Hole vs contradiction). A contradiction is a derivation of $\perp$ from the axioms. A hole is a desired coupling statement not derivable from the axioms; holes are promoted to conjectures or to new explicit axioms.

Remark 7.1 (Current explicit holes). (H1) *Zipper-to-holonomy*: Σ_{zip} determines Hol^{prot} (or forces $\text{Hol}^{\text{prot}} = 0$) under a natural rigidity hypothesis.

(H2) *Zipper algebra*: conditions under which $\triangleleft$ generates a monoid/semigroup action on S and induces functorial actions on invariants.

(H3) *Atlas-to-zipper*: $\text{Hol}^{\text{atlas}}$ is encoded by merge-tree data on I .

(H4) *Meta-closure tower*: explicit endofunctor M on charts capturing “closing the gap” while preserving soundness, and its (non-)fixed-point structure.

(H5) *Arithmetization layer*: minimal conditions under which internal certification implies a diagonal obstruction and hence $\kappa_R > 0$ for some targets.

(H6) *Interaction of bank-functor E and prediction-functor Pred_T* (a functorial bridge from certified parameter sets to prediction kernels).

7.1 Conjectures as Safe Network Expansions

Conjecture 7.1 (Zipper $\Rightarrow$ trivial protocol holonomy under rigidity). Assume a rigidity axiom identifying bank/margin refinements with homotopies that preserve the sublevel filtration on I and preserve MT_I . Then constancy of Σ_{zip} on I implies $\Delta_{\mathcal{Q}}(N, N', \tau, \delta; K) = 0$ for the induced window semantics.

Conjecture 7.2 (Zipper monoid and induced actions). Assume a set of teeth $\mathcal{T}$ equipped with a composition law $*$ such that applying τ_1 then τ_2 is equivalent (up to $\sim$) to applying $\tau_1 * \tau_2$. Then $\triangleleft$ defines a (partial) right action of a monoid $(\mathcal{T}, *)$ on S , and any functorial invariant of the induced filtrations is constant along action-orbits.

Conjecture 7.3 (Atlas holonomy as groupoid cohomology class). There exists a cohomology theory on the nerve $\text{Nerve}(\mathfrak{G}_r)$ in which $\text{Hol}^{\text{atlas}}$ is represented by a class of the cocycle $\{h_{ijk}\}$.

Conjecture 7.4 (Meta-closure tower has no sound complete fixed point). Assume a meta-operator M that extends a chart by adding internal certification power while preserving soundness. If M is sufficiently expressive to encode its own certification predicate, then there is no chart T such that $M(T) \cong T$ and certification becomes sound and complete at fixed finite resources.

Conjecture 7.5 (Zipper monoid action and complexity proxy). Let $\triangleleft$ be the zipper refinement operator (Axiom 3.3). There exists a monoid $(\mathcal{T}, *)$ and an action $\mathcal{T} \curvearrowright S$ such that each elementary refinement corresponds to multiplication by a generator in $\mathcal{T}$. Define the zipper complexity of reaching a target syntax token s^* from s as the minimal word length

$$\text{Comp}_{\triangleleft}(s \rightarrow s^*) := \min \left\{ \ell : \exists t_1 * \dots * t_\ell \in \mathcal{T} \text{ with } (t_1 * \dots * t_\ell) \cdot s = s^* \right\}.$$

In systems where refinements model coarse-graining, this length behaves analogously to circuit depth.

Remark 7.2 (Renormalization-group analogy). In QFT-style settings, a zipper step $\triangleleft$ admits an interpretation as integrating out degrees of freedom and producing an effective kernel. Non-commutativity of refinement programs corresponds to non-commutativity of coarse-graining steps under different scheme/order choices, with holonomy providing a quantitative invariant of the mismatch.

8 Network Diagram (Expansive)

Figure 1 summarizes the typed structure of the net: solid arrows are typed morphisms already defined in Sections 2–3, and dashed arrows labeled (H1)–(H2) are the declared coupling holes of Section 7.

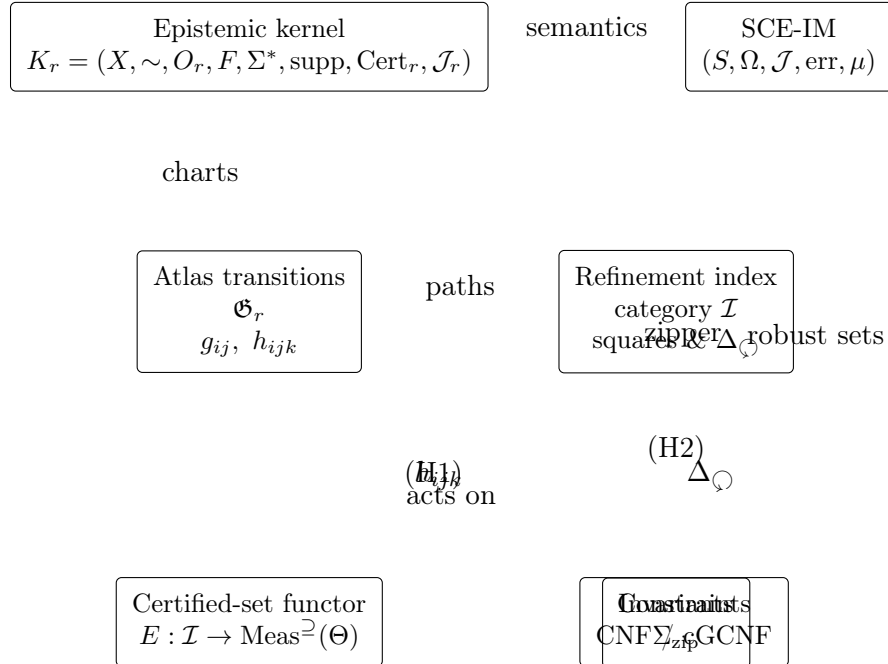


Figure 1: Closure net schematic. Solid arrows are typed morphisms already defined in Sections 2–3 (charts, refinements, and certified-set functors). Dashed arrows labeled (H1)–(H2) are declared coupling holes: they require additional hypotheses to connect zipper signatures to holonomy data. The diagram is intended as a navigation aid rather than a completeness claim.

9 Verifiable Extensions: Experimental Nodes and Protocols

9.1 Experimental Contract (Auditable Measurement)

Definition 9.1 (Verifiable extension). A verifiable extension adds a node D (data/harness) and a finite family of measurable observables $M = \{M_1, \dots, M_k\}$ together with:

- (V1) explicit domains (datasets or instance families) and windows K ,
- (V2) estimators $\widehat{M}_i$ computable from finite samples,
- (V3) auditable error bars (concentration bounds or bootstrap intervals),
- (V4) falsifiable predictions (inequalities or scaling laws with declared regimes).

Definition 9.2 (Window sampling oracle). An oracle for window sampling is a procedure returning i.i.d. samples $\theta_t \sim \mu_\Theta(\cdot \mid K)$ or MCMC samples with effective sample size $\text{ESS}(T)$. In this section, i.i.d. is treated as [Proved] for concentration bounds; MCMC bounds are treated as [Model].

9.2 Typed Protocol Holonomy: Deterministic vs History-Dependent Semantics

Remark 9.1 (Deterministic indexed sets imply trivial holonomy). If $E_\tau^{(N)}$ is defined as a function only of the terminal indices (N, τ) , then any two paths $(N, \tau) \rightarrow (N', \tau + \delta)$ with the same endpoint produce the same set and holonomy is forced to vanish. Therefore nontrivial protocol holonomy requires history-dependent refinement semantics.

Definition 9.3 (Refinement program category). Let $\mathcal{I}_{\text{prog}}$ be a category whose objects are certificate states

$$c = (S, \tau, \xi),$$

where S is a concrete bank instance (not just its size), τ is a margin parameter, and ξ is auxiliary state (random seed, optimizer state, stopping rule, etc.). Morphisms are program steps acting on c (bank augmentation, margin tightening, recomputation). A protocol is a path $p : c \rightarrow c'$ in $\mathcal{I}_{\text{prog}}$.

Definition 9.4 (Program-evaluated certified sets). A program-evaluated certification functor is a functor

$$E_{\text{prog}} : \mathcal{I}_{\text{prog}} \rightarrow \text{Meas}^\supset(\Theta), \quad E_{\text{prog}}(c) =: E_c \subseteq \Theta.$$

Two protocols $p_1, p_2 : c \rightarrow c_1, c_2$ may yield distinct outputs even if the terminal indices match (e.g. same $(N', \tau + \delta)$), because the terminal bank instance S or auxiliary state ξ can differ.

Definition 9.5 (Protocol holonomy observable (program semantics)). Fix a start state c and two protocols $p_1 : c \rightarrow c_1$ and $p_2 : c \rightarrow c_2$. For a window $K \subseteq \Theta$, define

$$\Delta_\circ(p_1, p_2; K) := \mu_\Theta((E_{p_1} \cap K) \triangle (E_{p_2} \cap K)),$$

where E_{p_i} denotes the output certified set $E_{\text{prog}}(c_i)$ produced by executing p_i from c (Definition 9.4). Definition 2.14 is recovered as a special case when certificate states are restricted to (N, τ) and programs are canonical.

Definition 9.6 (Normalized protocol holonomy). If $\mu_\Theta(K) > 0$, define the normalized magnitude

$$\tilde{\Delta}_\circ(p_1, p_2; K) := \frac{\Delta_\circ(p_1, p_2; K)}{\mu_\Theta(K)} \in [0, 1].$$

This removes dependence on the absolute scale of μ_Θ when only relative window fractions are desired.

9.3 Monte Carlo Estimators for Window-Measures

Definition 9.7 (Indicator estimator). Let $A \subseteq \Theta$ be measurable and let $\theta_t \sim \mu_\Theta(\cdot \mid K)$ i.i.d. Define

$$\hat{\mu}_T(A \cap K) := \mu_\Theta(K) \cdot \frac{1}{T} \sum_{t=1}^T \mathbf{1}_A(\theta_t).$$

Theorem 9.1 ([Proved] Hoeffding bound for window-measures). *Under i.i.d. sampling, for any $\epsilon > 0$,*

$$\Pr\left(|\hat{\mu}_T(A \cap K) - \mu_\Theta(A \cap K)| \geq \epsilon\right) \leq 2 \exp\left(-\frac{2T\epsilon^2}{\mu_\Theta(K)^2}\right).$$

Proof. Apply Hoeffding's inequality [5] to bounded random variables $X_t = \mu_\Theta(K) \mathbf{1}_A(\theta_t) \in [0, \mu_\Theta(K)]$. $\square$

Remark 9.2 ([Model] MCMC replacement). If only MCMC samples are available, replace T in Theorem 9.1 by $\text{ESS}(T)$ (using a declared estimator of integrated autocorrelation time). This substitution is exact for the variance of linear functionals under the ergodic-mean central-limit theorem; concentration-style tail bounds for general Markov chains additionally require mixing-rate control [6], and ESS is itself estimated, so the resulting bars are a [Model] contract that can be anti-conservative under slow mixing or nonlinear functionals.

9.4 Normalized Window Probabilities (Avoiding Explicit $\mu_\Theta(K)$)

Definition 9.8 (Conditional window measure). Assume $0 < \mu_\Theta(K) < \infty$. Define the conditional window probability measure

$$\mu_\Theta^K(A) := \Pr_{\theta \sim \mu_\Theta(\cdot \mid K)}[\theta \in A] = \frac{\mu_\Theta(A \cap K)}{\mu_\Theta(K)}.$$

Definition 9.9 (Normalized indicator estimator). Let $\theta_t \sim \mu_\Theta(\cdot \mid K)$ i.i.d. Define the normalized estimator

$$\hat{\mu}_T^K(A) := \frac{1}{T} \sum_{t=1}^T \mathbf{1}_A(\theta_t),$$

which estimates $\mu_\Theta^K(A)$ and does not require knowledge of $\mu_\Theta(K)$.

Theorem 9.2 ([Proved] Hoeffding bound for normalized window probabilities). *Under i.i.d. sampling, for any $\epsilon > 0$,*

$$\Pr\left(|\hat{\mu}_T^K(A) - \mu_\Theta^K(A)| \geq \epsilon\right) \leq 2 \exp(-2T\epsilon^2).$$

Proof. Apply Hoeffding's inequality [5] to Bernoulli random variables $\mathbf{1}_A(\theta_t) \in [0, 1]$. $\square$

Definition 9.10 (Normalized gray-zone and protocol-holonomy observables). For certified sets $E_1, E_2 \subseteq \Theta$, define the normalized protocol holonomy in window K :

$$\tilde{\Delta}_\square(E_1, E_2 \mid K) := \mu_\Theta^K(E_1 \triangle E_2).$$

For an ideal target region E_τ and a bank-certified region $E(S, \tau)$, define the normalized gray-zone as the normalized measure of the symmetric difference:

$$\tilde{G}(S, \tau \mid K) := \mu_\Theta^K(E_\tau \triangle E(S, \tau)).$$

Under the existential certification convention (Definition 9.15) one has $E(S, \tau) \subseteq E_\tau$, so this equals the certified-miss region $\mu_\Theta^K(E_\tau \setminus E(S, \tau))$; under the universal (sup) convention of the case studies (Section 10), where $E_\tau \subseteq E(S, \tau)$, it equals the false-positive region $\mu_\Theta^K(E(S, \tau) \setminus E_\tau)$. The symmetric difference is the approximation error of the bank-certified set with respect to the ideal set in either convention. Both observables are estimable by Definition 9.9 with error bars from Theorem 9.2.

Remark 9.3 (Absolute vs normalized measures). The absolute quantities Δ_Θ and G are recovered by multiplying by $\mu_\Theta(K)$. Normalized observables $\tilde{\Delta}_\Theta$ and $\tilde{G}$ are preferable when $\mu_\Theta(K)$ is unknown or when only conditional sampling is available.

9.5 Pre-Registration: Sample Size Planning and Decision Thresholds

Definition 9.11 (Sample size for normalized indicator tests). Fix accuracy $\epsilon \in (0, 1)$ and risk level $\eta \in (0, 1)$. For the normalized estimator in Definition 9.9, define

$$T(\epsilon, \eta) := \left\lceil \frac{1}{2\epsilon^2} \log \frac{2}{\eta} \right\rceil.$$

The validity of this choice is Proposition 9.1.

Proposition 9.1 ([Proved] Guaranteed confidence interval width). *If $T \geq T(\epsilon, \eta)$, then with probability at least $1 - \eta$,*

$$|\hat{\mu}_T^K(A) - \mu_\Theta^K(A)| \leq \epsilon.$$

Equivalently, the confidence interval $[\hat{\mu}_T^K(A) - \epsilon, \hat{\mu}_T^K(A) + \epsilon]$ is valid at level $1 - \eta$.

Proof. Apply Theorem 9.2 and solve $2 \exp(-2T\epsilon^2) \leq \eta$ for T . $\square$

Definition 9.12 (Decision rule schema). Given a pre-registered tolerance $\varepsilon_{\text{hol}} > 0$ and risk η , declare

$$\text{“holonomy detected”} \iff \hat{\Delta}_T - \epsilon \geq \varepsilon_{\text{hol}},$$

with ϵ chosen and $T \geq T(\epsilon, \eta)$ before sampling (A5). Analogous rules apply for gray-zone collapse tests $\tilde{G} \rightarrow 0$ by substituting $\tilde{G}_T$.

9.6 Distances Between Predicted Distributions (for Atlas Probes)

Definition 9.13 (Total variation and Wasserstein). For probability measures P, Q on $(Y, \mathcal{B})$:

$$d_{\text{TV}}(P, Q) := \sup_{A \in \mathcal{B}} |P(A) - Q(A)|.$$

If $Y \subseteq \mathbb{R}^m$ with metric $\|\cdot\|$, define the 1-Wasserstein distance (Kantorovich–Rubinstein dual form; [8])

$$W_1(P, Q) := \sup_{\text{Lip}(f) \leq 1} \left| \int f dP - \int f dQ \right|.$$

Definition 9.14 (Maximum mean discrepancy (MMD)). Fix a bounded positive definite kernel k on Y . Define

$$\text{MMD}_k^2(P, Q) := \mathbb{E} k(X, X') + \mathbb{E} k(Y, Y') - 2 \mathbb{E} k(X, Y),$$

with $X, X' \sim P$ i.i.d. and $Y, Y' \sim Q$ i.i.d. [7]. An unbiased empirical estimator is the U-statistic with self-pairs excluded: given samples $x_1, \dots, x_n \sim P$ and $y_1, \dots, y_m \sim Q$,

$$\widehat{\text{MMD}}_U^2(P, Q) = \frac{1}{n(n-1)} \sum_{i \neq j} k(x_i, x_j) + \frac{1}{m(m-1)} \sum_{i \neq j} k(y_i, y_j) - \frac{2}{nm} \sum_{i,j} k(x_i, y_j).$$

(Naive averages including diagonal self-kernel terms are biased.)

9.7 Experimental Nodes (P1–P5) as Network Expansions

9.7.1 P1: Refinement-Square Holonomy Experiment (Finite-Bank)

Definition 9.15 (Template-pool instantiation). Fix a large finite pool S_{pool} , serving as an empirical stand-in for an idealized infinite bank. For each template $s \in S_{\text{pool}}$, fix a measurable score function $F_s : \Theta \rightarrow \mathbb{R}$. For a concrete bank instance $S \subseteq S_{\text{pool}}$ and margin $\tau > 0$, define the certified set

$$E(S, \tau) := \left\{ \theta \in \Theta : \min_{s \in S} F_s(\theta) < -\tau \right\},$$

i.e., θ is certified when *some* bank template passes at margin τ (existential certification, matching the convention of Theorem 4.1). The ideal region is $E_\tau := E(S_{\text{pool}}, \tau)$, and $E(S, \tau) \subseteq E_\tau$ for every $S \subseteq S_{\text{pool}}$.

Lemma 9.1 ([Model] Pool Lipschitz-net transfer inclusions). *Assume the template pool S_{pool} is equipped with a metric d_S and that, for all $\theta \in K$, the score map is Lipschitz in the template index:*

$$|F_s(\theta) - F_{s'}(\theta)| \leq L d_S(s, s') \quad \forall s, s' \in S_{\text{pool}}.$$

If $S \subseteq S_{\text{pool}}$ is an ε -net of S_{pool} (every s is within ε of some $s' \in S$), then for all $\tau > 0$,

$$E(S_{\text{pool}}, \tau + L\varepsilon) \cap K \subseteq E(S, \tau) \cap K \subseteq E(S_{\text{pool}}, \tau) \cap K.$$

Thus, on the window K , the abstract transfer axiom $E_{\tau+\omega(\varepsilon)} \subseteq E_\tau^{(N)} \subseteq E_\tau$ holds in the pool model with $\omega(\varepsilon) = L\varepsilon$ (which is all the windowed estimators of Section 9 measure).

Proof. For $\theta \in K$: since $S \subseteq S_{\text{pool}}$, $\min_{s \in S_{\text{pool}}} F_s(\theta) \leq \min_{s \in S} F_s(\theta)$, which gives $E(S, \tau) \cap K \subseteq E(S_{\text{pool}}, \tau) \cap K$. For the first inclusion, take $\theta \in E(S_{\text{pool}}, \tau + L\varepsilon) \cap K$ and let $s_0 \in S_{\text{pool}}$ attain the pool minimum at θ (the pool is finite). Then $F_{s_0}(\theta) < -(\tau + L\varepsilon)$. By the ε -net property there is $s' \in S$ with $d_S(s_0, s') \leq \varepsilon$, and by Lipschitzness (valid for $\theta \in K$)

$$F_{s'}(\theta) \leq F_{s_0}(\theta) + L d_S(s_0, s') \leq F_{s_0}(\theta) + L\varepsilon < -\tau,$$

so $\min_{s \in S} F_s(\theta) < -\tau$, i.e. $\theta \in E(S, \tau) \cap K$. Taking $E_\tau^{(N)} := E(S, \tau)$ and $E_\tau := E(S_{\text{pool}}, \tau)$ recovers exactly the abstract transfer axiom of Theorem 4.1 on the window K , with $\omega(\varepsilon) = L\varepsilon$. $\square$

9.7.2 P1*: Proof-of-Concept Artifact Node (Cut-State Holonomy)

This node instantiates the paper’s “uncertainty $\rightarrow 0$ while $\Delta_\odot \not\rightarrow 0$ ” regime in a controlled, stylized lattice-extrapolation harness designed to mirror common workflow choices (fit windows / cuts) in numerical field theory.

Definition 9.16 (Lattice-extrapolation harness (stylized but auditable)). Let (a, m) range over a finite design grid (lattice spacing a and mass m). Observations are scalar summaries

$$y(a, m) = \theta_0 + c a^2 + f a^4 + d m + e a^2 m + \epsilon, \quad \epsilon \sim N(0, \sigma^2 / n_{\text{cfg}}),$$

with unknown target θ_0 and finite-configuration noise controlled by n_{cfg} . Define the prediction set for θ_0 as an interval $I_p(N)$ produced by a protocol p (below) with an auditable half-width estimator $\hat{w}_p(N)$.

n_{cfg}	N	$\hat{U}$	s.e.	$\hat{\Delta}_{\odot}$	s.e.
30	360	0.7992	0.009378	0.3009	0.009712
120	1440	0.4085	0.004582	0.1734	0.005332
480	5760	0.2045	0.002272	0.08338	0.002614
1920	23040	0.1014	0.001128	0.05201	0.001489
7680	92160	0.05144	0.0005736	0.04491	0.0009879
30720	368640	0.02628	0.0002840	0.04525	0.0005170

Table 1: Artifact node (same data as Figure 2). Here $N = 12n_{\text{cfg}}$ (twelve design points). The column $\hat{U}$ is the within-protocol half-width proxy (Definition 9.18) and $\hat{\Delta}_{\odot}$ is the protocol-holonomy estimator. Values are replication means over $R = 600$ independent Monte Carlo replications with fixed seed; the s.e. columns are the replication standard deviations divided by $\sqrt{R}$. Reproduced by `artifact/P1star_cut_state/p1star_cut_state.py`.

Definition 9.17 (Two protocols distinguished by a cut-state ξ). Fix a design grid with four a -values and three m -values. Protocol p_{fine} uses only the two finest a -values in the $a^2 \rightarrow 0$ regression stage (whereas p_{full} uses all four). Both complete the same terminal extrapolation to $(a, m) = (0, 0)$, but with different internal state

$$\xi \in \{\text{fine cut}, \text{full fit}\}.$$

Let $I_{\text{fine}}(N)$ and $I_{\text{full}}(N)$ denote the resulting intervals.

Definition 9.18 (Uncertainty-width and holonomy estimators). With $u = a^2$, each protocol mid-point is a fixed linear functional ϕ_p of the 12 design observations (per- m ordinary least squares followed by the exact $m \rightarrow 0$ stage). Within one Monte Carlo replication define the pooled variance estimate $\hat{\sigma}^2 := \text{RSS}/(12 - 5)$ from the full fit, the intercept standard errors $s.e._p := \hat{\sigma} \|\phi_p\|$ (the per-observation noise variance is σ^2/n_{cfg} , so $\hat{\sigma}^2$ already carries the n_{cfg} scaling), and the half-widths $\hat{w}_p(N) := 2 s.e._p$. Then a within-protocol uncertainty proxy (not the global gray-zone set-measure in Definition 9.22) is

$$\hat{U}(N) := \max\{\hat{w}_{\text{fine}}(N), \hat{w}_{\text{full}}(N)\}, \quad \hat{\Delta}_{\odot}(N) := |\text{mid}_{\text{fine}}(N) - \text{mid}_{\text{full}}(N)|,$$

the midpoint separation of the two prediction intervals $I_p(N) := \text{mid}_p(N) \pm \hat{w}_p(N)$. (The Hausdorff distance of the two intervals is $\text{dist}_H(I_{\text{fine}}, I_{\text{full}}) = |\text{mid}_{\text{fine}} - \text{mid}_{\text{full}}| + |\hat{w}_{\text{fine}} - \hat{w}_{\text{full}}|$; since both half-widths vanish as $N \rightarrow \infty$ by Proposition 9.2, the Hausdorff distance and the midpoint separation share the same limit $|f|u_1u_2$, and the node reports the midpoint separation.) Across R independent replications the point estimates are the replication means and the reported standard errors are the replication standard deviations divided by $\sqrt{R}$.

Proposition 9.2 ([Model] Persistent cut-holonomy diagnostic). *For the harness in Definition 9.16 on the grid of Definition 9.17 (with $0 < u_1 < u_2$ and all four u -values distinct), as $n_{\text{cfg}} \rightarrow \infty$ (limits in probability),*

$$\text{mid}_{\text{fine}} \rightarrow \theta_0 - f u_1 u_2, \quad \text{mid}_{\text{full}} \rightarrow \theta_0,$$

and consequently

$$\lim_{N \rightarrow \infty} \hat{\Delta}_{\odot}(N) = |f| u_1 u_2, \quad \lim_{N \rightarrow \infty} \hat{U}(N) = 0.$$

Thus, for $f \neq 0$, $\hat{U} \rightarrow 0$ while $\hat{\Delta}_{\odot} \not\rightarrow 0$: the persistent separation is exactly the fine-cut extrapolation bias, and it flags the hidden-state dependence (ξ encodes the fit window).

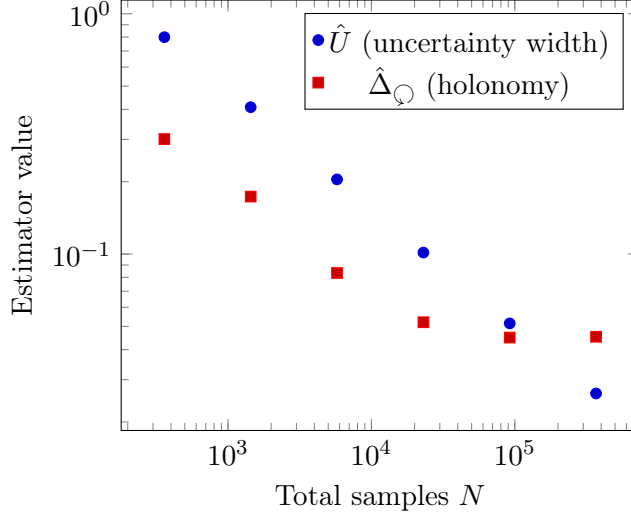


Figure 2: Artifact node: estimated uncertainty width $\hat{U}$ and protocol-holonomy $\hat{\Delta}_{\mathcal{Q}}$ vs total samples N (log–log axes). The design uses two protocol states $\xi \in \{\text{fine cut, full fit}\}$; the limiting separation equals the model-predicted value $|f| u_1 u_2 = 0.045$ (Proposition 9.2), reflecting model curvature ($f a^4$) and window choice. Error bars are replication standard errors (Table 1).

Proof. Per fixed m , write $g(u, m) = \theta_0 + dm + (c + em)u + fu^2$. The fine protocol fits the line through $(u_1, g(u_1, m))$ and $(u_2, g(u_2, m))$; its intercept at $u = 0$ is

$$\frac{u_2 g(u_1, m) - u_1 g(u_2, m)}{u_2 - u_1} = \theta_0 + dm - f u_1 u_2$$

(the $(c+em)$ -terms cancel). This is affine in m , so the $m \rightarrow 0$ stage is exact and $\text{mid}_{\text{fine}} = \theta_0 - f u_1 u_2$ up to zero-mean noise vanishing with n_{cfg} . The full protocol fits the exact model $\{1, u, u^2\}$ per m (the four distinct u -values make it identifiable), so $\text{mid}_{\text{full}} \rightarrow \theta_0$. Hence $\hat{\Delta}_{\mathcal{Q}}(N) = |\text{mid}_{\text{fine}} - \text{mid}_{\text{full}}| \rightarrow |f| u_1 u_2$, and $\hat{U}(N) \rightarrow 0$ since $\hat{\sigma} \rightarrow \sigma / \sqrt{n_{\text{cfg}}} \rightarrow 0$. $\square$

Remark 9.4 (Concrete numbers). The harness parameters and the decision rule below (Definition 9.19) were frozen before sampling (A5) and are recorded in the artifact audit log (Remark 9.5); the numbers that follow report the pre-registered outcome. For the frozen design grid $a \in \{0.1, 0.1225, 0.2, 0.4\}$, $m \in \{-0.5, 0, 0.5\}$, $\theta_0 = 1$, $c = -2$, $d = 1.5$, $e = 0.7$, $f = 299.875$ (so that $|f| u_1 u_2 = 0.045$), $\sigma = 1.119$, $R = 600$ replications and fixed seed, Monte Carlo evaluation produces the trend in Figure 2 and Table 1 (generated by `artifact/P1star_cut_state/p1star_cut_state.py`). In this instance $\hat{U}$ decays as $N^{-1/2}$ (fitted log–log slope -0.4944 ± 0.0019 , 1σ propagated from the replication standard errors) while $\hat{\Delta}_{\mathcal{Q}}$ approaches the model-predicted constant $|f| u_1 u_2 = 0.045$ (Table 1: $0.3009 \rightarrow 0.0453$). Both pre-registered decisions of Definition 9.19 pass: collapse ($\hat{U} = 0.0263 \leq \varepsilon_U = 0.05$) and persistent holonomy ($\hat{\Delta}_{\mathcal{Q}} - 2 \text{ s.e.} = 0.0442 \geq \Delta_{\min} = 0.0225$).

Definition 9.19 (Pre-registered decision for the artifact node). Frozen before sampling (A5): the harness parameters of Remark 9.4, the estimators of Definition 9.18, and the decision rule at the largest N :

- (1) *Uncertainty collapse*: declare “collapse” iff $\hat{U}(N_{\max}) \leq \varepsilon_U := 0.05$.
- (2) *Persistent holonomy*: with model-predicted separation $\Delta_{\infty} = |f| u_1 u_2$, test Null : $\Delta_{\infty} = 0$ against Alt : $\Delta_{\infty} \geq \Delta_{\min} := |f| u_1 u_2 / 2$; accept Alt iff $\hat{\Delta}_{\mathcal{Q}}(N_{\max}) - 2 \text{ s.e.}(\hat{\Delta}_{\mathcal{Q}}) \geq \Delta_{\min}$.

The node claim (C6) is declared falsified if collapse fails or Alt is not accepted.

Remark 9.5 (Reproducibility). The full source code that generates Table 1 and Figure 2, including the frozen parameters, random seeds and audit logs, accompanies this manuscript in the repository directory `artifact/P1star_cut_state/` (script `p1star_cut_state.py`, results `table_results.csv`, audit `audit_log.json`); Figure 2 is rendered from the same data with `pgfplots`. The audit log satisfies the artifact-bundle contract R1–R5 of Definition 10.8 (synthetic data generator, code, seed log, parameter record, and report with the pre-registered decisions).

9.7.3 P1^{**}: Public-Data Instantiation (GW150914 Skeleton)

[*Protocol*] This optional node upgrades P1^{*} from a stylized harness to a public-data instantiation, aligning with Case Study C (Definition 10.7) and the GW150914 release [11]. We pre-register two analysis states $\xi \in \{\xi_{\text{bandpass}}, \xi_{\text{whiten}}\}$ (detrending/whitening/PSD windowing) and define a square by swapping the order of whitening and bandpass filtering. The artifact bundle includes a runnable script (`artifact/P2_gw150914_skeleton/gw150914_skeleton.py`) that downloads calibrated strain segments from GWOSC and computes an auditable parameter estimate (a Newtonian chirp-mass proxy) under both protocol orders, together with the resulting holonomy estimator $\hat{\Delta}_{\mathbb{Q}}$; the script refuses to run until a pre-registration file (A5) is frozen.

Definition 9.20 (Adaptive bank augmentation operator). Fix a window K and an exploration distribution ν supported on K . Given (S, τ) , define an augmentation step Aug by selecting the template with maximal expected margin violation at the *current* margin,

$$s^* \in \arg \max_{s \in S_{\text{pool}} \setminus S} \mathbb{E}_{\theta \sim \nu} \left[\max(F_s(\theta) + \tau, 0) \right],$$

(ties broken by a fixed pre-registered order on S_{pool}), then setting

$$S \leftarrow S \cup \{s^*\}.$$

Because the selection objective depends on τ , augmentation and margin tightening generically do not commute; this makes bank refinement history-dependent, enabling nontrivial holonomy (Remark 9.1).

Definition 9.21 (Program square (two orders)). Define two protocols from the same start state $c_0 = (S, \tau, \xi)$:

$$p_1 := \text{Aug}^m \circ \text{Tighten}_{\delta}, \quad p_2 := \text{Tighten}_{\delta} \circ \text{Aug}^m,$$

where Aug^m denotes m augmentation steps (targeting bank size increase) and Tighten_{δ} updates $\tau \leftarrow \tau + \delta$ (possibly also re-evaluating certificates). Define $\Delta_{\mathbb{Q}}(p_1, p_2; K)$ by Definition 9.5.

Remark 9.6 (Measurable test target). In the pool instantiation, $\Delta_{\mathbb{Q}}(p_1, p_2; K)$ reduces to the window measure of a symmetric difference between two explicit sets $E(S_1, \tau_1)$ and $E(S_2, \tau_2)$, and is estimable via Theorem 9.1.

Estimator pseudocode (auditable).

Input: start state $c_0 = (S, \tau, \xi)$, protocols p_1, p_2 , window K , sample size T

- 1) Execute p_1 from c_0 to obtain certified set E_1 (membership oracle: $\theta \rightarrow \mathbf{1}_{E_1}(\theta)$)
- 2) Execute p_2 from c_0 to obtain certified set E_2 (membership oracle)
- 3) Sample $\theta_1, \dots, \theta_T \sim \mu_{\Theta}(\cdot \mid K)$ (i.i.d. or MCMC with ESS)
- 4) Return $\mu(K) \cdot \frac{1}{T} \sum_t \mathbf{1}[(\theta_t \in E_1) \text{ XOR } (\theta_t \in E_2)]$

Output: $\hat{\Delta}_{\mathbb{Q}}$ and Hoeffding error bar (or ESS-adjusted bar)

9.7.4 P2: Gray-Zone Scaling (Estimating Effective Dimension)

Definition 9.22 (Gray-zone observable). Assume an ideal region E_τ is available in the pool model, for instance $E_\tau := E(S_{\text{pool}}, \tau)$. For a finite bank instance S (size N), define the gray-zone in window K as the certified-miss region

$$G(S, \tau; K) := \mu_\Theta((E_\tau \setminus E(S, \tau)) \cap K) = \mu_\Theta((E_\tau \triangle E(S, \tau)) \cap K),$$

the equality holding because $E(S, \tau) \subseteq E_\tau$ under the existential convention of Definition 9.15. The normalized gray zone of Definition 9.10 is $\tilde{G}(S, \tau | K) = G(S, \tau; K)/\mu_\Theta(K)$.

Definition 9.23 (Empirical covering dimension). Let (S_{pool}, d_S) be the template pool metric space, with S_{pool} the pool of Definition 9.15 and d_S the pool metric of Lemma 9.1. For $\varepsilon > 0$ define the covering number

$$N(\varepsilon) := \min \left\{ m : \exists s_1, \dots, s_m \in S_{\text{pool}} \text{ with } S_{\text{pool}} \subseteq \bigcup_{j=1}^m B_\varepsilon(s_j) \right\}.$$

Define the effective covering dimension

$$d_{\text{eff}} := \limsup_{\varepsilon \downarrow 0} \frac{\log N(\varepsilon)}{\log(1/\varepsilon)}.$$

For a single fixed finite pool this limsup is 0 (eventually $N(\varepsilon) = |S_{\text{pool}}|$), so the asymptotic law of Proposition 9.3 must be read in the two-scale regime $N \ll |S_{\text{pool}}|$. Operationally, a two-scale estimator uses $\varepsilon_1 < \varepsilon_2$ and

$$\hat{d} = \frac{\log N(\varepsilon_1) - \log N(\varepsilon_2)}{\log(\varepsilon_2/\varepsilon_1)}.$$

Proposition 9.3 ([Model] Log-log slope test). Assume (i) a Hölder-type transfer modulus $\omega(\varepsilon) \approx L\varepsilon^\alpha$ on window K ; (ii) covering control $\varepsilon_N \asymp N^{-1/d_{\text{eff}}}$ along the bank sweep, with d_{eff} estimated as in Definition 9.23 (two-scale regime); and (iii) band regularity: the pushforward of μ_Θ^K by the pool-minimum score $F^*(\theta) := \min_{s \in S_{\text{pool}}} F_s(\theta)$ has a density bounded between c and C on a fixed neighborhood $[-\tau - t_0, -\tau]$ of the level $-\tau$ (so that

$$\mu_\Theta^K(\{\theta : -(\tau + t) \leq F^*(\theta) < -\tau\}) \asymp t \quad \text{for } 0 \leq t \leq t_0).$$

Then, for N large enough that $\omega(\varepsilon_N) \leq t_0$,

$$G(N, \tau; K) \leq C \mu_\Theta(K) \omega(\varepsilon_N) \asymp N^{-\alpha/d_{\text{eff}}}.$$

If, in addition, (iv) boundary resolution: on a μ_Θ^K -comparable fraction of the band $\{-(\tau + \omega(\varepsilon_N)) \leq F^* < -\tau\}$ the pool minimizer $s^*(\theta) := \arg \min_{s \in S_{\text{pool}}} F_s(\theta)$ is resolved by the bank only at covering scale, $\text{dist}_S(s^*(\theta), S) \geq c_0 \varepsilon_N$, with the lower sensitivity bound $F_s(\theta) \geq F^*(\theta) + c_1 \text{dist}_S(s, s^*(\theta))$ for all $s \in S_{\text{pool}}$, and with $c_0 c_1 > L$ when $\alpha = 1$, then the matching lower bound holds and

$$G(N, \tau; K) \asymp N^{-\alpha/d_{\text{eff}}}.$$

The slope α/d_{eff} is estimable by linear regression on $\log \hat{G}$ vs $\log N$ with uncertainty from Theorem 9.1, together with a separate estimate of d_{eff} from pool covering numbers; conditions (iii)–(iv) are themselves pre-registered empirical conditions of the P2 node (Remark 9.9).

Derivation sketch. By Lemma 9.1 (existential form) $E_{\tau+\omega(\varepsilon_N)} \cap K \subseteq E(S, \tau) \cap K \subseteq E_\tau \cap K$, so $E_\tau \setminus E(S, \tau) \subseteq \{\theta \in K : -(\tau + \omega(\varepsilon_N)) \leq F^*(\theta) < -\tau\}$; band regularity then gives the upper bound $G \leq C \mu_\Theta(K) \omega(\varepsilon_N) \asymp N^{-\alpha/d_{\text{eff}}}$. For the lower bound, take θ in the comparable fraction of the band granted by (iv): $F^*(\theta) < -\tau$ puts $\theta \in E_\tau$, while for every $s \in S$,

$$F_s(\theta) \geq F^*(\theta) + c_1 \text{dist}_S(s, s^*(\theta)) \geq -(\tau + \omega(\varepsilon_N)) + c_0 c_1 \varepsilon_N,$$

which is $\geq -\tau$ for $c_0 c_1 \varepsilon_N \geq \omega(\varepsilon_N) \approx L \varepsilon_N^\alpha$ (automatic for $\alpha < 1$ as $\varepsilon_N \rightarrow 0$; by hypothesis for $\alpha = 1$); hence $\theta \notin E(S, \tau)$, so the gray zone contains a μ_Θ^K -comparable fraction of the band and $G \gtrsim \mu_\Theta(K) \omega(\varepsilon_N)$. $\square$

9.7.5 P3: Atlas Holonomy with Probes (Groupoid on Kernels)

Definition 9.24 (Kernel action on measures). Let $(Y, \mathcal{B})$ be measurable and let K be a Markov kernel on Y . Let Prob denote the category whose objects are measurable spaces and whose morphisms are Markov kernels. For a probability measure $\nu \in \text{Prob}(Y)$ define its pushforward by K as

$$K_\# \nu(A) := \int_Y K(A \mid y) d\nu(y), \quad A \in \mathcal{B}.$$

If K is an isomorphism in Prob , then $K_\# : \text{Prob}(Y) \rightarrow \text{Prob}(Y)$ is bijective.

Definition 9.25 (Probe-set holonomy magnitude (section + operator tests)). Fix a finite probe set $\{o_\ell\}_{\ell=1}^L \subseteq O_r$ and a distance d on $\text{Prob}(Y_{o_\ell})$ (e.g. d_{TV} , W_1 , or MMD). Let h_{ijk, o_ℓ} denote the component of the atlas cocycle acting on $\text{Prob}(Y_{o_\ell})$ and interpret its action by pushforward (Definition 9.24).

(Section-based magnitude) Define

$$\Delta_{ijk}^{\text{sec}} := \sum_{\ell=1}^L d\left(\text{Pred}_{T_i}(o_\ell), (h_{ijk, o_\ell})_\# \text{Pred}_{T_i}(o_\ell)\right).$$

(Operator-detecting magnitude) Fix, for each ℓ , a finite family of test measures $\{\nu_{\ell, m}\}_{m=1}^{M_\ell} \subseteq \text{Prob}(Y_{o_\ell})$ (“probe measures”). Define

$$\Delta_{ijk}^{\text{op}} := \sum_{\ell=1}^L \sum_{m=1}^{M_\ell} d\left(\nu_{\ell, m}, (h_{ijk, o_\ell})_\# \nu_{\ell, m}\right).$$

Finally set $\Delta_{ijk}^{\text{atlas}} := \Delta_{ijk}^{\text{sec}} + \Delta_{ijk}^{\text{op}}$. The operator test detects holonomy even when the particular predicted section $\text{Pred}_{T_i}(o_\ell)$ is fixed by h_{ijk, o_ℓ} , *provided the probe family separates the identity from h_{ijk, o_ℓ}* (i.e., for each component with $h_{ijk, o_\ell} \neq \text{id}$ there is a probe moved by it); in general the inference is one-directional: $\Delta_{ijk}^{\text{atlas}} > 0$ certifies nontrivial holonomy, while a vanishing value does not exclude it (probe-blindness is recorded as a declared hypothesis).

Empirical estimator (samples). For each probe o_ℓ , obtain sample batches from $\text{Pred}_{T_i}(o_\ell)$ and from $(h_{ijk} \cdot \text{Pred}_{T_i})(o_\ell)$ by simulating kernels. Estimate d by empirical TV/Wasserstein/MMD; attach bootstrap confidence intervals.

9.7.6 P4: Zipper vs Protocol Holonomy Correlation Test (Bridge H1 as an Experiment)

Definition 9.26 (Operational correlation observable). Fix a family of resource settings producing zipper signatures $\Sigma_{\text{zip}}(E, o)$ and protocol holonomy magnitudes $\Delta_\circ$. Define the correlation

functional

$$\text{Corr} := \text{corr}\left(\mathbf{1}[\Sigma_{\text{zip}} \text{ stable on } I], \Delta_{\odot}\right),$$

where stability is defined by a declared tolerance in κ , merge-tree edit distance on MT_I , and hitting-time deviation. Each $\Delta_{\odot}$ is estimated via Theorem 9.1; estimator errors attenuate the correlation and must be propagated (bootstrap over the resource settings).

Remark 9.7 (Falsifiable bridge). Conjecture 7.1 becomes an experimentally checkable implication: “stability of Σ_{zip} ” $\Rightarrow$ “ $\Delta_{\odot} \approx 0$ ” in the tested regime.

9.7.7 P5: CNF-to-Volume Calibration Node

Definition 9.27 (Discrete-to-continuous embedding calibration). For a discrete CNF F on $\{0, 1\}^n$, embed assignments into $[0, 1]^n$ by mapping each vertex to its unit cube cell. Let $K = [0, 1]^n$ and let μ be Lebesgue measure. Then $\text{Vol}_K(\text{Mod}(F))$ equals $\#\text{SAT}(F) \cdot 2^{-n}$ exactly in this embedding.

Remark 9.8 (Purpose). This node provides a sanity check for Monte Carlo volume estimators and connects the CNF/cGCNF node to the window-volume node.

9.8 Pre-Registered Falsification Patterns (Diagnostics)

Definition 9.28 (Falsification pattern). A falsification pattern for an experimental node P is a pre-registered triple $\Pi_P^f = (\text{estimator}, \text{tolerance}, \text{failure event})$ such that the node’s model claim is declared refuted at confidence $1 - \eta$ whenever the failure event occurs under the frozen harness (A5). A falsification of a node refutes the node’s claim, not the net: the response is a typed network expansion in a separate post-mortem branch (Remark 9.11).

Remark 9.9 (Pre-registered falsification patterns per node). (P1) Failure event: $\Delta_{\odot}(p_1, p_2; K) \leq \varepsilon_{\text{hol}}$, i.e. no detectable holonomy at the pre-registered tolerance ε_{hol} of Definition 9.12 under the decision rule of Definition 9.30, although the τ -dependent augment rule of Definition 9.20 predicts non-commutativity; interpretation: the augment rule fails to couple to the margin in the tested harness.

(P1*) Failure event: $\hat{U}(N_{\text{max}}) > \varepsilon_U$ (no uncertainty collapse) or $\hat{\Delta}_{\odot}(N_{\text{max}}) - 2 \text{ s.e.} < \Delta_{\min}$ (no persistent separation); interpretation: misspecified noise model or the two protocols are equivalent in the tested regime (Definition 9.19).

(P2) Failure event: the fitted log-log slope of $\log \hat{G}$ vs $\log N$ deviates from α/d_{eff} beyond the propagated error bars; interpretation: transfer-modulus or band-regularity misspecification (Proposition 9.3).

(P3) Failure event: $\Delta_{ijk}^{\text{atlas}} \leq \varepsilon$ at the probe resolution while an independent operator certificate shows $h_{ijk} \neq \text{id}$; interpretation: probe-blindness (the separating condition of Definition 9.25 fails), recorded as a new hypothesis.

(P4) Failure event: Σ_{zip} -stability coexists with $\Delta_{\odot} \geq \varepsilon_{\text{hol}}$ beyond error bars; interpretation: Conjecture 7.1 fails in the tested regime (the rigidity hypothesis is not met).

(P5) Failure event: the Monte Carlo estimate $\widehat{\text{Vol}}_K(\text{Mod}(F))$ differs from $\#\text{SAT}(F) \cdot 2^{-n}$ beyond the Monte Carlo error; interpretation: estimator or embedding bug (the exact identity of Definition 9.27 is an arithmetic check).

9.9 Pre-Registered Protocols (Decision Rules with Error Bars)

Definition 9.29 (Pre-registration schema). For each experiment $P \in \{P1, \dots, P5\}$, a pre-registration is the tuple

$$\Pi_P = (\text{Input}, \text{Output}, \widehat{\text{Est}}, \text{ErrBar}, \text{Null}, \text{Alt}, \text{Decision}),$$

where:

- Input fixes the harness (window K , distribution μ_Θ , pool/banks, probes),
- Output names the target observable(s) $(G, \Delta_\circlearrowleft, \Delta^{\text{atlas}})$,
- $\widehat{\text{Est}}$ specifies the estimator (e.g. Theorem 9.1),
- ErrBar specifies an auditable error bar (Hoeffding or ESS-adjusted),
- Null and Alt are mutually exclusive inequalities to test,
- Decision is a threshold rule with declared confidence.

Definition 9.30 (Example: P1 commutation test). Fix start state c_0 , two protocols p_1, p_2 (Definition 9.21), window K , and sample size T . Let $\hat{\Delta}_T$ be the Monte Carlo estimator of $\Delta_\circlearrowleft(p_1, p_2; K)$. Declare error bar ϵ_T such that

$$\Pr(|\hat{\Delta}_T - \Delta_\circlearrowleft| \geq \epsilon_T) \leq \eta.$$

A concrete choice under i.i.d. is

$$\epsilon_T := \mu_\Theta(K) \sqrt{\frac{\log(2/\eta)}{2T}} \quad (\text{Theorem 9.1}).$$

Define:

$$\text{Null} : \Delta_\circlearrowleft(p_1, p_2; K) = 0, \quad \text{Alt} : \Delta_\circlearrowleft(p_1, p_2; K) \geq \Delta_{\min}.$$

Decision rule: accept Alt if $\hat{\Delta}_T - \epsilon_T \geq \Delta_{\min}$; otherwise do not reject Null.

Remark 9.10 (Additional pre-registrations). Analogous pre-registrations are defined for P2 (log–log slope test), P3 (atlas operator test using probe measures $\nu_{\ell, m}$), P4 (correlation threshold), and P5 (CNF-to-volume exactness check).

Definition 9.31 (Holonomy–gap diagnostic). In the finite-bank harness, measure both G and $\Delta_\circlearrowleft$ along a bank-sweep $N \uparrow$. Declare:

$$(D1) \quad G \rightarrow 0 \wedge \Delta_\circlearrowleft \not\rightarrow 0 \implies \text{additional non-commutativity source (new node/edge required)}.$$

Remark 9.11 (Network-expansive response rule). When (D1) occurs, extend the net by adding an explicit node capturing the extra source (e.g. optimizer state, discretization, numerical tolerances), and re-type refinements as morphisms acting on that node. The rule is a pre-registered post-mortem expansion (A5): it preserves consistency while expanding explanatory power, and it does not rescue the refuted node claim, which remains recorded as falsified.

10 Physics/Tech Case Studies (End-to-End Harness Instantiations)

Remark 10.1 (Case studies are instantiation templates). The case studies in this section are schemas mapping standard scientific pipelines into the closure-net types. They are not presented as new physics claims; their role is to specify exactly what would be measured (certified sets, gray zones, holonomy magnitudes) under a frozen, pre-registered experimental node.

10.1 Case Study A: Materials Stability Under Competing Phases

Definition 10.1 (Materials harness). Let Θ be a measurable parameter space of material candidates (e.g. compositions, structures, thermodynamic settings), with window $K \subseteq \Theta$ and sampling measure $\mu_\Theta(\cdot \mid K)$. Let S_{pool} be a large pool of competing phases or competing decompositions. For each $s \in S_{\text{pool}}$, define a measurable violation score

$$F_s(\theta) := \Delta E_s(\theta),$$

interpreted as an energy-above-competitor quantity (lower is better). Fix a margin $\tau > 0$ (stability slack) and define the ideal stability region

$$E_\tau := \left\{ \theta \in \Theta : \sup_{s \in S_{\text{pool}}} F_s(\theta) \leq -\tau \right\}.$$

For a finite competitor-bank instance $S \subseteq S_{\text{pool}}$, define the finite-bank certified region

$$E(S, \tau) := \left\{ \theta \in \Theta : \sup_{s \in S} F_s(\theta) \leq -\tau \right\}.$$

The materials harness uses the *universal* certification convention (all bank templates must pass), which is the physically correct robustness notion for stability: $E_\tau \subseteq E(S, \tau)$ for every $S \subseteq S_{\text{pool}}$, bank densification shrinks the certified region, and the functor typing of Definition 2.13 is satisfied directly (with the gray zone given by the symmetric difference of Definition 9.10, here the false-positive region $E(S, \tau) \setminus E_\tau$).

Definition 10.2 (Materials refinement programs). A certificate state is $c = (S, \tau, \xi)$ where ξ records the selection rule, random seeds, surrogate-model state, and stopping criteria. Define:

(A1) Tighten step $\text{Tighten}_\delta : (S, \tau, \xi) \mapsto (S, \tau + \delta, \xi)$.

(A2) Augment step $\text{Aug} : (S, \tau, \xi) \mapsto (S \cup \{s^*\}, \tau, \xi')$ where

$$s^* \in \arg \max_{s \in S_{\text{pool}} \setminus S} \mathbb{E}_{\theta \sim \mu_\Theta(\cdot \mid K)} \left[\max(F_s(\theta) + \tau, 0) \right],$$

and ξ' updates the selection trace.

The augment rule is history-dependent through ξ and through the empirical objective approximated from finite samples; its τ -dependence makes the square of Definition 10.3 generically non-commutative.

Definition 10.3 (Pre-registered material protocols). Fix $m \in \mathbb{N}$ augment steps and $\delta > 0$. From a common start $c_0 = (S_0, \tau_0, \xi_0)$ define two protocols:

$$p_1 := \text{Aug}^m \circ \text{Tighten}_\delta, \quad p_2 := \text{Tighten}_\delta \circ \text{Aug}^m.$$

Let E_{p_i} be the resulting certified set in Θ produced by executing p_i . Define normalized holonomy and gray-zone observables using Definition 9.10:

$$\tilde{\Delta}_\circ(p_1, p_2 \mid K) := \mu_\Theta^K(E_{p_1} \triangle E_{p_2}), \quad \tilde{G}(S, \tau \mid K) := \mu_\Theta^K(E_\tau \triangle E(S, \tau)).$$

Proposition 10.1 ([Model] Materials holonomy–gap diagnostic). *In the materials harness, perform a bank sweep (increasing $|S|$) while recording $\tilde{G}$ and $\tilde{\Delta}_\circ$. If $\tilde{G} \rightarrow 0$ but $\tilde{\Delta}_\circ \not\rightarrow 0$, then the observed non-commutativity is not explained by finite competitor coverage alone and must be attributed to additional program state ξ (e.g. surrogate bias, adaptive sampling, numerical tolerances), which is represented by a new node/edge in the closure net.*

10.2 Case Study B: Collider Analysis Pipeline

Definition 10.4 (Collider harness). Let D denote a dataset of binned observables (histograms) and let Θ be a measurable parameter space of interest (signal strengths, EFT coefficients, nuisance parameters). Fix a window $K \subseteq \Theta$ and sampling measure $\mu_\Theta(\cdot \mid K)$. Let S_{pool} be a pool of systematic variations and simulation configurations. For each $s \in S_{\text{pool}}$ and analysis configuration ξ (cuts, calibrations, unfolding choices), define a measurable discrepancy score

$$F_s(\theta; \xi) := \chi^2(D, \hat{D}_s(\theta; \xi)),$$

where $\hat{D}_s(\theta; \xi)$ is the predicted binned distribution under variation s . For a finite bank $S \subseteq S_{\text{pool}}$ and tolerance $\tau > 0$, define the certified region

$$E(S, \tau, \xi) := \left\{ \theta \in \Theta : \sup_{s \in S} F_s(\theta; \xi) \leq \tau \right\}.$$

As in Case Study A the convention is universal (all variations must agree), and $E_\tau := E(S_{\text{pool}}, \tau, \xi) \subseteq E(S, \tau, \xi)$.

Definition 10.5 (Pipeline refinements as programs). A certificate state is $c = (S, \tau, \xi)$ where ξ includes: cut thresholds, calibration constants, unfolding regularization, and random seeds. Define:

(B1) Tighten step $\text{Tighten}_\delta : (S, \tau, \xi) \mapsto (S, \tau - \delta, \xi)$ (stricter agreement), or alternatively $\xi \mapsto \xi'$ as stricter cuts.

(B2) Augment step $\text{Aug} : (S, \tau, \xi) \mapsto (S \cup \{s^*\}, \tau, \xi')$ selecting

$$s^* \in \arg \max_{s \in S_{\text{pool}} \setminus S} \mathbb{E}_{\theta \sim \mu_\Theta(\cdot \mid K)} \left[\max(F_s(\theta; \xi) - \tau, 0) \right].$$

Non-commutativity arises because $F_s(\cdot; \xi)$ depends on ξ , and ξ changes under tightening. *Margin orientation:* here the margin enters as an upper tolerance ($\{\sup_s F_s \leq \tau\}$), so tightening *decreases* τ —the reverse of the bank-transfer convention where the margin grows ($\{F < -\tau\}$); the program-level holonomy formalism is agnostic to this orientation, only the monotonicity of the certified-set functor is required.

Definition 10.6 (Pre-registered collider holonomy test). Fix a start state $c_0 = (S_0, \tau_0, \xi_0)$ and two protocols p_1, p_2 (tighten-then-augment vs augment-then-tighten). Estimate $\tilde{\Delta}_\square(p_1, p_2 \mid K)$ by sampling $\theta_t \sim \mu_\Theta(\cdot \mid K)$ and computing

$$\hat{\Delta}_T = \frac{1}{T} \sum_{t=1}^T \mathbf{1}[\theta_t \in E_{p_1} \triangle E_{p_2}],$$

with Hoeffding error bars from Theorem 9.2. Declare holonomy-detection at confidence $1 - \eta$ if the lower confidence bound exceeds a pre-registered tolerance $\varepsilon_{\text{hol}} > 0$.

10.3 Case Study C: Gravitational-Wave Coherence Across Detectors

Definition 10.7 (GW multi-detector coherence harness). Let $D = \{d_\alpha\}_{\alpha \in \{\text{H1, L1, V1, ...}\}}$ be calibrated strain time series from multiple detectors. Let Θ parameterize waveform families and nuisance degrees of freedom; fix window $K \subseteq \Theta$ and sampling $\mu_\Theta(\cdot \mid K)$. Let S_{pool} be a pool of stress

tests (glitch models, noise nonstationarity segments, calibration perturbations). For $s \in S_{\text{pool}}$ and analysis state ξ (PSD estimation, windowing, vetoes), define

$$F_s(\theta; \xi) := \max_{\alpha} \text{Res}(d_{\alpha}, h_{\alpha}(\cdot; \theta); s, \xi),$$

a measurable residual statistic (e.g. normalized energy in the whitened residual, or negative log-likelihood gap). Define the certified region

$$E(S, \tau, \xi) := \left\{ \theta : \sup_{s \in S} F_s(\theta; \xi) \leq \tau \right\}.$$

Bank augmentation adds stress tests; tightening increases coherence requirements by decreasing τ or strengthening vetoes in ξ (margin orientation as in Definition 10.5). The holonomy observable $\tilde{\Delta}_{\odot}(p_1, p_2 \mid K)$ is defined as in Definition 9.10.

Remark 10.2 (Utility). Case Studies A–C instantiate the same abstract observables $(\tilde{G}, \tilde{\Delta}_{\odot})$ in three domains. They operationalize “network expansion” as a diagnostic tool: persistent holonomy after gray-zone collapse forces an explicit new node/edge for the missing mechanism. Pre-registration templates are given by Definitions 10.3 (case A) and 10.6 (case B); cases C and D adopt the same schema via Definition 9.29, with their declared $(K, \mu_{\Theta}(\cdot \mid K), \tau, \xi, \varepsilon_{\text{hol}}, \eta)$ frozen before sampling (A5).

10.4 Reproducibility Bundle (Artifact Node)

Definition 10.8 (Artifact bundle). A reproducibility artifact bundle for a case study is a tuple

$$\text{Art} = (\text{DataID}, \text{CodeID}, \text{SeedLog}, \text{Params}, \text{Report}),$$

where:

- (R1) DataID identifies the dataset snapshot (DOI/run list/hash) and preprocessing recipe.
- (R2) CodeID identifies code version (commit hash/container digest) implementing membership oracles for $E(\cdot)$, protocols, and estimators.
- (R3) SeedLog records all randomness: seeds, initial banks S_0 , and augmentation traces.
- (R4) Params records $(K, \mu_{\Theta}(\cdot \mid K), \tau_0, \delta, m, \varepsilon_{\text{hol}}, \eta)$ and the distance choice for atlas probes.
- (R5) Report contains $\hat{G}$ and $\hat{\Delta}$ with confidence intervals and the pre-registered decisions (Definition 9.12).

Remark 10.3 (Minimal public data anchors). The case studies admit concrete public anchors without changing the formalism:

- (A1) Materials: a public DFT repository snapshot plus a fixed competitor-pool extraction rule for S_{pool} .
- (A2) Collider: a public HEP open-data release plus a fixed histogramming + simulation/systematics pool definition.
- (A3) GW: a public calibrated strain release plus a fixed residual statistic and a fixed stress-test pool definition.

The manuscript treats these anchors as parameters of Art; theorems depend only on measurability and the sampling contract.

10.5 Case Study D: Lattice QCD Systematics as Protocol Holonomy (Instantiation Template)

Let Θ parameterize a lattice ensemble family (lattice spacing a , box size L , quark masses, and algorithmic tolerances). Let a “bank” S_N be a finite set of gauge configurations / solver initializations / multistart heuristics used to certify a target observable, e.g. a hadron mass estimator $M_p(\theta)$ with a robustness margin τ on residuals or fit quality.

Two refinement protocols. A typical non-commuting square interchanges the two extrapolation orders:

p_1 : increase statistics/bank $\Rightarrow$ continuum extrapolation $\Rightarrow$ chiral extrapolation,

p_2 : increase statistics/bank $\Rightarrow$ chiral extrapolation $\Rightarrow$ continuum extrapolation.

Execute both as program paths in $\mathcal{I}_{\text{prog}}$ with fixed, pre-registered auxiliary state ξ (seeds, fit priors, stopping rules), and compare the resulting certified parameter sets or certified output intervals via $\tilde{\Delta}_{\square}$. A nonzero value quantifies systematic path dependence under finite resources, in the same operational sense as in the finite-bank instantiation of Section 9.7.

Falsifiable prediction skeleton. Under a transfer modulus $\omega(\varepsilon_N)$ and stable extrapolation maps, one expects $\tilde{\Delta}_{\square} \rightarrow 0$ as $N \rightarrow \infty$, with a rate controlled by $\omega(\varepsilon_N)$ and the effective dimension of the banked configuration/template space (Section 9.7.4).

A Provenance and Scope

- Added typed refinement category $\mathcal{I}$ and functor E ; replaced untyped inverse-commutator by square-comparison holonomy (Defs. 2.11–2.14). Bank densification is typed contravariantly on the index, matching the monotonicity of the transfer theorem (Theorem 4.1).
- Added prediction functors into the Markov-kernel category and typed atlas transitions as a cochain of bijections with triple-overlap cocycles (Defs. 2.16–2.20); the groupoid reading holds exactly when the cocycles are trivial, and the general datum is gerbe-like (Conjecture 7.3).
- Resolved notation collision by renaming the cGCNF ambient product space to $\mathcal{X}$ (no change in meaning), keeping X for the kernel state space of Definition 2.2.
- Standardized the bank-certification convention: the pool model (Definition 9.15) and the case studies are typed consistently (existential certification in the pool model; universal certification in the case studies, with the gray zone measured by the symmetric difference of Definition 9.10).
- Completed the pre-registration discipline: falsification patterns (Section 9.8), the P1* pre-registration (Definition 9.19), and the reproducibility artifacts (**artifact**/).
- Extended holes list to include the missing bridge between bank-certified parameter sets and prediction kernels.

B Bibliography (Minimal Pointers)

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